

Market-Oriented Industrial Policy and Labor Demand: Evidence from Government Guidance Funds in China*

Wenzhe Zhao[†]

School of Economics, CUFE

Man Zhu[‡]

School of Economics, CUFE

This version: April 2026

Abstract

This paper studies how government venture capital shapes labor demand. We combine verified Government Guidance Fund (GGF) investment records with firm-level online vacancy data to estimate direct effects on treated firms, spillovers through observed supply-chain links, and city-industry reallocation patterns in a staggered event-study design. Our main GGF-centered results show that GGF investment raises hiring and posting at treated firms. CSMAR top-five customer and supplier links also exhibit positive labor-demand responses, while coarser same-city and input-output exposure designs do not support a broad positive local-spillover interpretation. Government-procurement evidence points to a demand-support channel along GGF-linked production networks. Vacancy-based city-industry evidence likewise indicates that post-exposure labor-demand growth is concentrated in targeted firms and industries rather than diffusing uniformly across local economies. A private-VC comparison is used only as a benchmark for distinctiveness and external validity, not as a clean mechanism test. Auxiliary tax evidence is supportive only: wage-per-worker dynamics are directionally consistent, but employment pre-trends fail in the matched tax sample. Taken together, the results imply that policy-targeted equity capital primarily reallocates labor demand toward selected firms and linked production networks rather than generating broad-based local job creation.

JEL Classification: G28; J23; O25

Keywords: Government Guidance Funds; Industrial Policy; Labor Demand

*We are grateful to colleagues and seminar participants for helpful comments and discussions. All errors remain our own.

[†]School of Economics, Central University of Finance and Economics. Email: wenzhe.zhao@cufe.edu.cn.

[‡]Corresponding author. School of Economics, Central University of Finance and Economics. Email: man.zhu@ontariotechu.net.

1. Introduction

Government-directed venture capital has become a central instrument of modern industrial policy, especially in China, where Government Guidance Funds (GGFs) channel large volumes of state-backed equity into strategic sectors (Juhász et al., 2024). Yet its labor-market footprint remains poorly understood. Does GGF-backed investment simply expand treated firms, or does it also reallocate labor demand across linked firms, rival firms, sectors, and local markets? This paper studies that question by combining verified GGF investment records with firm-level online vacancy data. Our main results are GGF-centered: GGF-backed firms expand hiring and postings; CSMAR top-five customer and supplier links also expand; coarser same-city and IO-based exposure designs do not show broad positive spillovers; and city-industry evidence points to concentrated reallocation rather than broad job creation. We use private VC later as a benchmark for distinctiveness and external validity, but not as a co-equal organizing treatment arm.

The economic distinction between GGF and private VC is not that one is public and the other private; it is that they operate under different objective functions. Private VC is disciplined primarily by financial returns. GGF capital is deployed under a dual mandate that combines return-seeking with explicit policy goals such as sector upgrading, strategic resilience, and local development. This distinction is visible in recent evidence showing that government VC and private VC select different firms and respond differently to geopolitical shocks (Chen et al., 2025). In labor-market terms, the implication is straightforward: if GGF capital is more concentrated in targeted firms, sectors, and places than private VC, then labor-demand responses should differ not only at treated firms but also across local rivals and neighboring markets.

Our empirical approach follows directly from this question. We construct a matched panel of GGF-backed firms, private VC-backed firms, and untreated firms by combining verified fund-investment records with vacancy postings from a major Chinese recruitment platform. Vacancies allow us to observe labor demand directly: hiring quantity, posting quantity, posted wages, and the skill composition of vacancies (Deming and Kahn, 2018; Hershbein and Kahn,

2018; Acemoglu et al., 2022). We then estimate staggered event studies for both direct effects and local spillovers. Direct estimates compare treated firms to untreated controls over time. Spillover estimates focus on never-treated firms exposed to investment shocks in the same city $\times$ industry market, allowing us to distinguish direct firm expansion from broader equilibrium adjustment in local labor markets.

Three results organize the paper. First, GGF investment raises hiring and posting at treated firms and is accompanied by more selective wage and education upgrading. Second, indirect effects depend sharply on the exposure margin. CSMAR top-five customer and supplier links exhibit positive labor-demand responses, with customer links estimated more precisely, and procurement rises along GGF-linked own-firm and customer-link paths, consistent with a demand-support channel. By contrast, same-city $\times$ same-industry and IO-based BL/FL exposure designs do not support a broad positive local-multiplier interpretation. Third, broader city-industry evidence points to concentrated reallocation: post-exposure growth is concentrated in targeted firms, linked production networks, and selected industries rather than diffusing uniformly across local labor markets. The private-VC comparison is used later as a benchmark for how distinctive these GGF patterns are, while auxiliary tax and listed-firm evidence remain supportive rather than headline validation samples.

This paper contributes to three strands of literature. First, it contributes to the economics of industrial policy and labor demand. Criscuolo et al. (2019) show that targeted industrial support raises employment at subsidized firms; Kline and Moretti (2014) and Lane (2025) emphasize that place-based and sectoral industrial policy can reshape local economies through reallocation and network propagation; and Rotemberg (2019) shows that concentrated firm subsidies can reduce gains at rivals through equilibrium adjustment. We bring these insights to the setting of government venture capital and show that the relevant equilibrium margin is not only whether treated firms expand, but whether labor demand is reallocated toward targeted firms and sectors.

Second, we contribute to the literature on venture capital and government-backed finance. Work on private VC shows that venture capital accelerates firm professionalization, investment efficiency, and growth (Hellmann and Puri, 2002; Chemmanur et al., 2011; Bernstein

et al., 2016; Puri and Zarutskie, 2012). Work on public financing shows that state-backed capital can relax financing constraints and alter firm growth paths (Lerner, 1999; Howell, 2017). Our closest conceptual benchmark is Chen et al. (2025), who show that government VC differs from private VC precisely because it follows a different investment mandate. We extend that distinction to labor markets: the contribution here is not to claim a clean horse race in which GGF uniformly dominates private VC, but to show that a private-VC benchmark does not fully subsume the GGF labor-demand patterns we document in the main analysis.

Third, we contribute to the literature using vacancy data to study firm behavior and labor-market structure. Vacancy postings have become a central measure of firm-level labor demand because they reveal hiring quantity, posted wages, and skill requirements at high frequency (Deming and Kahn, 2018; Hershbein and Kahn, 2018; Acemoglu et al., 2022). We use those margins to show that the labor-market footprint of government venture capital extends beyond simple job counts. The evidence is most consistent with stronger direct recruitment growth at targeted firms, selective wage and education upgrading, demand propagation along CSMAR top-five customer and supplier links, and limited broad local spillovers.

Finally, we demonstrate the value of high-frequency vacancy microdata for evaluating industrial policy in near real time. Job-posting data have been used to study skill demand across firms and labor markets (Deming and Kahn, 2018; Acemoglu et al., 2022), cyclical skill adjustment (Hershbein and Kahn, 2018), and macro shocks (Forsythe et al., 2020). We show that vacancy microdata can identify the effects of a discrete policy event—with treatment timing established from verified investment records—and that posting-based effects can be benchmarked against narrower auxiliary datasets without making those auxiliary samples carry the main identification burden.

The remainder of the paper is organized as follows. Section 2 provides institutional background on GGFs. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 reports direct effects on invested firms. Section 6 presents CSMAR top-five customer and supplier spillovers, IO (BL/FL) incidence, distributional/composition incidence, mechanism evidence, and the VC benchmark. Section 7 presents GE / aggregate reallocation evidence. Section 8 clarifies scope and auxiliary validation. Section 9 concludes.

2. Institutional Context

2.1. Government Guidance Funds as Market-Oriented Industrial Policy

Government Guidance Funds are equity vehicles sponsored by central or local governments and co-financed by private investors. The model formalizes a public “anchor” tranche that is intended to crowd in private capital and delegate investment decisions to professional fund managers. National guidance for venture-capital guidance funds was issued in 2008 and a formal regulatory framework was promulgated by the Ministry of Finance in 2015, which standardized governance, risk-sharing, and reporting requirements (State Council General Office, 2008; Ministry of Finance, 2015). In practice, GGFs have targeted early- and growth-stage private firms in priority sectors—advanced manufacturing, semiconductors, clean energy, biomedicine, and AI—while largely excluding large listed or state-owned enterprises. Relative to conventional subsidy tools, this design is closer to government-backed venture capital programs studied in Lerner (1999), Brander et al. (2015), Grilli and Murtinu (2014), and, in the recent China context, Chen et al. (2025).

The scale of the program is substantial. By 2021 more than 1,800 GGFs had been established with target capital exceeding US\$1.5 trillion. The rapid expansion coincides with China’s broader pivot toward market-oriented industrial policy, highlighting the need to evaluate outcomes beyond investment flows and innovation metrics (Juhász et al., 2024; Criscuolo et al., 2019; Kalouptsidi, 2018).

Figure 1 plots the annual number of newly established GGFs (left axis) and cumulative committed capital (right axis) from 2008 to 2023, highlighting the post-2015 acceleration. The vertical marker at 2015 corresponds to the Ministry of Finance’s formal regulatory framework. The post-2018 moderation in committed capital is consistent with broader deleveraging and asset-management tightening during that period, which affected private-equity fundraising conditions.

[Insert Figure 1 here]

2.2. Why Labor Demand May Respond

GGF investment can affect treated firms’ labor demand through three direct channels. First, equity financing can relax binding financial constraints and enable scale expansion, increasing hiring across the skill distribution (Chodorow-Reich, 2014; Howell, 2017). Second, government-backed investment can act as a certification signal, raising the firm’s attractiveness to skilled workers and partners (Hellmann and Puri, 2002; Chemmanur et al., 2011; Bernstein et al., 2016). Third, the policy embeddedness of GGF managers may facilitate access to local procurement, regulatory approvals, and talent programs, amplifying labor demand beyond pure capital infusion (Howell, 2020; Giannetti et al., 2021).

These direct-effect channels also imply spillovers. A local demand expansion at treated firms can tighten labor markets, raise wages at competing employers, and reallocate workers across firms and cities (Moretti, 2010; Rotemberg, 2019; Berger et al., 2022; Azar et al., 2024; Moretti and Wilson, 2017). Whether GGF activity generates net new employment or primarily reshapes its distribution is an empirical question our design directly addresses (Busso et al., 2013; Kline and Moretti, 2014; Rotemberg, 2019; Lu et al., 2019; Autor et al., 2014).

3. Data and Sample Construction

3.1. GGF Investment Records

Treatment timing comes from cleaned venture-capital investment records built from Zero2IPO and matched to a verified government-venture-capital universe. Following recent work on government VC (Chen et al., 2025), we combine official GGF lists, fund-level business-registration information, and investment-event records to identify funds with clear government-guidance characteristics and then construct firm-level investment histories. For each investee, we identify the first GGF investment year and aggregate investment counts and committed capital. For firm-level spillovers, we no longer define exposure using the first GGF investment anywhere in a city. Instead, the baseline spillover treatment is the first GGF investment in the same city $\times$ mapped industry as the never-treated firm, which better matches the relevant local labor market for crowd-out and reallocation.

3.2. Online Job Postings (2016–2024)

We measure labor demand using vacancy microdata from a major Chinese online recruitment platform. Each posting reports the firm name, job location, education requirement, number of hires, and posted wage bounds. We aggregate postings to the firm-year level and construct six core outcomes: (i) log hiring count, (ii) log posting count, (iii) log mean maximum wage, (iv) log mean minimum wage, (v) high-skill share (bachelor’s degree or above), and (vi) technical share (engineering, software, R&D). This follows the vacancy-based measurement approach in [Deming and Kahn \(2018\)](#), [Hershbein and Kahn \(2018\)](#), [Acemoglu et al. \(2022\)](#), and [Azar et al. \(2024\)](#), allowing us to jointly track quantity, price, and composition of labor demand. Vacancy outcomes should be interpreted as intended labor demand rather than realized hires; a key empirical risk is platform-margin adoption (firms posting more on this platform after GGF investment). We therefore treat vacancy estimates as demand signals and triangulate with tax-based employment and wage outcomes from an independent administrative source.

3.3. Tax-Survey Outcomes (2007–2020)

We complement vacancy outcomes with administrative tax-survey data covering 2007–2020. The tax sample is independent of the vacancy data and includes treated and never-treated firms. We use it only for auxiliary outcomes—most usefully wage-per-worker and selected balance-sheet variables—rather than as a second headline employment design. Because the matched tax employment series exhibits pre-trend violations, the tax evidence is treated as supportive and directional rather than as a clean validation sample. Appendix Table [A9](#) reports the relative coverage of the recruitment, tax, and listed-firm auxiliary samples.

3.4. CSMAR Top-Five Customer and Supplier Links

To measure observed production-network exposure, we use CSMAR top-five customer and supplier records. These records identify major reported counterparties of listed firms and allow us to classify untreated counterparties as customers or suppliers of GGF-backed firms. We define the linked exposure year as the first year in which an untreated counterparty is connected through a CSMAR top-five customer or supplier relationship to a GGF-backed firm.

This design is narrower than city-industry exposure, but it more directly captures propagation along observed transaction links.

3.5. Firm Matching and Sample Construction

We harmonize firm names using a standardized cleaning routine that removes legal suffixes and punctuation and, where available, merge firms to unified registration identifiers (USCC or tax ID). We then match GGF investees to vacancy firms in two steps: (i) exact matching on cleaned firm name and city (to reduce false positives), and (ii) exact matching on cleaned firm name for remaining unmatched cases. Ambiguous name–city matches are flagged and excluded from treatment assignment. This conservative strategy prioritizes precision over recall.

The vacancy panel covers 2016–2024 and contains 1,764,759 observations and 1,006,764 unique firms. Our merged treated sample includes 2,515 unique firms across 29 first-investment cohorts (1994–2024). Because many cohorts fall outside the vacancy window or lack a full pre-period observation, the event-study and descriptive-statistics exhibits use smaller effective treated samples, such as the 603 firms observed at $t = -1$ in Table 1. Treated firms nevertheless represent only about 0.25% of unique firms in the panel, reflecting narrow policy targeting toward strategic sectors. The spillover sample consists of never-treated firms whose own city-industry market is eventually exposed to a GGF-backed investee, which yields a smaller but more targeted comparison group than the legacy city-level spillover design.

3.6. Descriptive Statistics

Figure 1 maps the city-level distribution of investees, demonstrating national coverage beyond a few core cities. Table 1 reports summary statistics in the pre-treatment period ($t = -1$ for treated firms): treated firms are larger recruiters and post higher wages before investment, consistent with positive selection into GGF funding. These baseline differences motivate our within-firm event-study design rather than cross-sectional comparisons. The appendix coverage diagnostics show that the vacancy sample remains the broadest and most policy-relevant

labor-demand sample available for this paper, while tax and listed-firm samples are narrower and therefore reserved for auxiliary analysis.

[Insert Figure 1 here]

[Insert Table 1 here]

4. Identification Strategy

GGF fund managers select firms with high growth potential, so cross-sectional comparisons confound treatment effects with selection. We address this using within-firm variation in treatment timing in a staggered difference-in-differences design, complemented by event-study dynamics.

4.1. Invested-Firm Dynamics

Our baseline event-study specification is:

$$Y_{it} = \sum_{k \neq -1} \beta_k \cdot \mathbb{I}(\text{rel_year}_{it} = k) + \mu_i + \lambda_{pt} + \varepsilon_{it}, \quad (1)$$

where Y_{it} is an outcome for firm i in year t and $\text{rel_year}_{it} = t - t_0(i)$ is years since first GGF investment. We bin event time at the endpoints: $k = -4$ represents all years ≤ -4 and $k = 3$ represents all years ≥ 3 . The omitted category is $k = -1$. Firm fixed effects μ_i absorb time-invariant differences, and province-by-year fixed effects λ_{pt} absorb regional shocks. Standard errors are clustered at the firm level. The pre-endpoint bin ($k = -4$) should be interpreted as an average pre-period test pooling all years ≤ -4 , rather than a year-by-year test of flatness far from treatment.

For average effects, we summarize post-investment outcomes with a single post-treatment indicator in the same fixed-effects framework.

4.2. Non-Invested Firms in Exposed Cities

For spillovers, we restrict to firms that never receive GGF investment and ask whether their outcomes change after GGFs begin operating in their city. We define $t_0^{\text{city}}(c)$ as the first year in which any firm in city c receives GGF investment, and estimate:

$$Y_{ict} = \sum_{k \neq -1} \delta_k \cdot \mathbb{1}(\text{rel_spill}_{ct} = k) + \mu_i + \lambda_{pt} + u_{ict}, \quad (2)$$

where $\text{rel_spill}_{ct} = t - t_0^{\text{city}}(c)$, firm fixed effects μ_i absorb time-invariant differences, and province-by-year fixed effects λ_{pt} absorb regional shocks while preserving within-province, cross-city variation in GGF exposure. We cluster standard errors at the city level to account for common exposure timing within cities. Event time is binned symmetrically at $k = -4$ and $k = 3$. We do not include city-by-year fixed effects in this spillover specification because they would absorb the city-year treatment variation that identifies city-level exposure timing.

4.3. Threats to Identification

The key identifying assumption is parallel trends: absent GGF investment (or city-level exposure), treated and control firms would have evolved similarly. We assess this assumption through pre-treatment coefficients in the event-study graphs. A related concern is anticipation: firms may accelerate hiring after term sheets are signed but before investment is formally recorded. We assess this by checking whether the normalized baseline period around $t = -1$ exhibits systematic upward movement relative to $t = -2$ and $t = -3$ in Figure 2; we do not observe a distinct anticipatory jump in the direct-effect outcomes.

Remaining threats include correlated city-level policy bundles (for example, concurrent local innovation or talent programs) and platform-margin measurement in vacancy data. Province-by-year fixed effects absorb broad regional shocks, and our spillover sample restriction to never-treated firms isolates indirect city-level exposure; however, city-specific policy confounders may still remain. We therefore interpret spillover magnitudes with caution and complement vacancy-based evidence with tax and city-level outcomes.

A separate concern is platform-margin adoption: GGF investees may begin using the recruitment platform more intensively after receiving institutional backing, even without a comparable increase in underlying hiring demand. We address this in two ways. First, auxiliary tax outcomes move in the same direction as the vacancy data, although we do not treat the tax employment estimate as a clean validation because the matched employment series exhibits pre-trend violations. Second, we report three harmonized vacancy estimates for $\ln(\text{hiring})$: 0.290 (SE 0.080) in the current Figure 2 headline sample, 0.189 (SE 0.042) in the broader platform full panel, and 0.134 (SE 0.053) in the pre-existing-user subsample. These estimates differ because they target different sample scopes, windows/cohort restrictions, and estimands (headline direct effect versus platform-margin decomposition), not because of a contradiction in sign. We therefore keep the Figure 2 estimate as the headline direct effect and use the platform decomposition to bound potential platform-adoption inflation.

5. Direct Effects on Treated Firms

5.1. Labor Demand Responses: Quantity, Wages, and Skill Composition

We begin with the labor-demand response at GGF-backed firms. The central question in this section is whether GGF investment is followed by a clear expansion in firm-level labor demand, and if so, along which margins. We focus first on the headline hiring and posting outcomes, then turn to posted wages and vacancy composition. Throughout, the emphasis is on the vacancy-based response of treated firms themselves rather than on broader equilibrium incidence, which we take up in the next section.

[Insert Figure 2 here]

Figure 2 presents the event-study evidence for the main direct-effect margins. The most pronounced and stable responses are on hiring and posting quantity. In both cases, the pre-treatment coefficients are relatively flat, while the post-treatment coefficients turn positive on impact and remain elevated thereafter. This pattern is consistent with a discrete upward shift in treated firms' labor demand around the first GGF investment rather than with a continuation

of pre-existing growth trends. The same figure also shows more selective positive movement in posted wages and in the high-education share of vacancies. By contrast, the technology-oriented vacancy margin is visibly weaker and should be read as supplementary rather than as a central direct effect.

[Insert Table 2 here]

Table 2 summarizes average treatment effects estimated by replacing the event-study dummies with a single post-investment indicator in the same fixed-effects framework. The quantity margins are the clearest. GGF investment raises log hiring by 0.290 (SE 0.080) and log postings by 0.265 (SE 0.061), implying sizable increases in firm-level recruitment activity after treatment. These are economically meaningful effects and line up closely with the event-study profiles in Figure 2.

The wage and composition responses are also positive, but somewhat more selective. Posted maximum wages rise by 0.105 log points (SE 0.063), while the minimum-wage estimate is 0.091 log points (SE 0.060). Taken together, the wage columns point to an upward shift in posted pay, although the evidence is stronger at the upper bound than at the lower bound. On composition, the high-education share increases by 0.053 (SE 0.022), indicating that treated firms tilt vacancy demand toward more educated workers after GGF investment. The technology-posting margin, however, is close to zero at -0.007 (SE 0.017). We therefore do not frame the direct effect as a broad increase in all forms of skill intensity. The stronger conclusion is narrower and more credible: GGF expands recruitment activity and is accompanied by higher posted wages and a greater demand for college-educated labor.

The direct-spec reconciliation table clarifies the 0.189-versus-0.290 discrepancy. In the current Figure 2 headline setup, the $\ln(\text{hiring})$ ATE is 0.290 (SE 0.080). In the broader platform-margin full panel, the corresponding estimate is 0.189 (SE 0.042). Restricting treated firms to pre-existing platform users yields 0.134 (SE 0.053). The gap is therefore mechanical rather than contradictory: the three estimates come from different sample scopes, event-window/cohort restrictions, and design targets (headline Figure 2 direct effect versus platform-margin decom-

position). We keep 0.290 as the main Figure 2 headline and use 0.189 and 0.134 as decomposition benchmarks.

An additional external validation based on listed-firm employee records points in the same direction on quantity. In the matched listed-firm subsample, the post-GGF coefficient for log total employees is 0.171 (SE 0.039), while the college-share outcome is close to zero and imprecise. We therefore use this external panel as support for quantity expansion at treated firms, but do not over-interpret it as strong composition evidence.

The combination of Figure 2 and Table 2 thus points to a fairly disciplined direct-effect pattern. The first-order response is expansion in hiring and posting. That quantity expansion is accompanied by higher posted pay and by a shift toward more educated vacancy profiles. At the same time, the weaker technology-posting result cautions against overstating how broad the composition response is. The direct evidence supports a story of stronger labor demand at treated firms, with upgrading on selected margins, rather than a uniform increase across every skill indicator.

6. Supply-Chain Spillovers and Mechanisms

6.1. CSMAR Top-Five Customer and Supplier Spillovers

Our primary spillover design is built on CSMAR top-five customer and supplier records. We classify untreated counterparties into customer and supplier roles relative to GGF-backed firms and then estimate event studies around the first linked exposure year. This design directly tracks propagation through observed transaction networks rather than through coarse city-level exposure cells.

[Insert Figure 4 here]

[Insert Table 3 here]

The CSMAR top-five counterparty evidence is positive on both customer and supplier margins, with customer links estimated more precisely. On the customer side, pre-treatment co-

efficients are close to zero and statistically flat, while post-treatment outcomes move upward in the preferred ATE summary: log hiring rises by 0.330 (SE 0.173), log postings by 0.213 (SE 0.129), and posted wages by 0.108–0.120 log points (SE 0.069–0.075). Supplier quantity estimates are also positive, at 0.319 (SE 0.371) for log hiring and 0.317 (SE 0.255) for log postings, but are less precise because the matched supplier sample is smaller. We therefore frame the CSMAR top-five customer and supplier evidence as the core observed-network spillover object, while distinguishing the stronger precision of the customer estimates from the more suggestive supplier estimates.

6.2. IO-Based BL/FL Spillovers

We next report the IO-based BL/FL exercise using city-industry mapping and linked treatment timing at the BL/FL margin. Unlike the CSMAR top-five counterparty design above, this IO design captures broad upstream/downstream exposure at the industry level. The two designs answer related but distinct questions: firm-to-firm demand propagation versus wider production-network adjustment.

[Insert Figure 3 here]

[Insert Table 4 here]

In the current preferred specification, BL and FL quantity margins are negative post-treatment: BL is -0.231 (SE 0.083) for log hiring and -0.203 (SE 0.077) for log postings, while FL is -0.183 (SE 0.047) and -0.176 (SE 0.048). We therefore do not present the IO taxonomy block as additional positive spillover evidence. Instead, we read it as incidence heterogeneity across linkage types under a coarser industry-mapping design, with the CSMAR top-five customer and supplier estimates retained as the main observed-network spillover object. The same-city $\times$ same-industry spillover figure is reported in the appendix as a robustness visual rather than as a main-text object.

6.3. Distributional Incidence and Composition Change

After documenting direct effects and supply-chain spillovers, we next examine who adjusts most within exposed markets. Table 5 provides a compact heterogeneity view by splitting the sample into pre-exposure incumbents and size groups. The purpose is not to create a new identification design, but to clarify distributional incidence around the main treatment and spillover estimates.

Two findings are most relevant. First, composition-share responses are modest in magnitude: in the listed panel, BA+ share is 0.004 (SE 0.010), and in the v6 composition panel the high-education and technical posting shares are 0.008 (SE 0.012) and -0.021 (SE 0.010). Second, the listed-panel quantity anchor is positive at 0.168 (SE 0.048). We therefore interpret Table 5 as descriptive incidence evidence on where adjustment appears stronger, rather than as a separate causal object.

6.4. Mechanism: Government Procurement

To explain why GGF-linked supply chains expand labor demand, we test a government-procurement mechanism. The empirical question is whether procurement contracts and contract amounts rise along GGF-related supply-chain exposure margins, and whether the corresponding VC-linked margins are weaker.

[Insert Figure 5 here]

[Insert Table 6 here]

The main-text mechanism table focuses on GGF-linked own-firm and customer-link transmission. In the post-ATE summary, own-firm GGF procurement rises by 0.314 (SE 0.052) in cumulative amount and 0.078 (SE 0.016) in cumulative contract count; customer-linked GGF exposure yields larger increases of 0.993 (SE 0.204) and 0.259 (SE 0.062). This pattern supports the channel interpretation that government procurement provides demand support to GGF-linked production networks, which then transmits to labor demand outcomes among CSMAR top-five customer and supplier counterparties.

6.5. VC Benchmark within the Spillover Block

We place the private-VC comparison inside the spillover section as a benchmark rather than as a separate headline object. The benchmark asks whether the supply-chain labor-demand propagation documented above is a generic VC pattern or a GGF-specific one.

[Insert Table 7 here]

In the spillover benchmark, both GGF and private-VC on-impact coefficients are small and statistically insignificant on quantity outcomes. For private VC, the event-year-0 coefficients are -0.011 (SE 0.020) for log hiring and -0.006 (SE 0.016) for log postings. We therefore read private VC as a calibration benchmark rather than as a competing main design.

7. GE / Aggregate Concentrated Expansion

Sections 5 and 6 document a firm-level pattern of direct expansion, supply-chain propagation, and heterogeneous incidence under GGF exposure. The remaining aggregate question is how those micro effects map into city-industry-level concentrated expansion. Section 7 therefore moves to the GE layer and asks whether post-exposure labor-demand growth is diffuse across local industries or concentrated in targeted segments. The baseline GE specification uses the updated symmetric window-3 event-time design ($k \in [-3, 3]$) to align with the current main-table release.

We do so using a refreshed city-year aggregate panel that combines two reproducible labor-demand sources: (i) vacancy-side recruitment outcomes from the latest v6 chain and (ii) listed-firm employee structure outcomes from the validated matching chain. City exposure timing is anchored to the first GGF-backed listed-firm arrival in that city, and regressions include city and year fixed effects.

[Insert Figure 8 here]

Figure 8 plots the event-time path for the two headline aggregate quantity outcomes (v6 ln postings and listed ln employees). The figure is designed as a compact GE diagnostic for

whether post-exposure changes appear in observable labor-demand aggregates without reverting to old city-total-employment measures.

[Insert Table 8 here]

Table 8 reports the same refreshed GE object in a compact cross-source format: quantity and structure margins for recruitment and listed employee aggregates side by side. The post effects on quantity are 0.091 (SE 0.108) in the recruitment aggregate and 0.190 (SE 0.139) in the listed-employee aggregate. The corresponding structure-share effects are -0.007 (SE 0.018) and -0.039 (SE 0.017). We therefore frame the GE evidence as suggestive concentrated expansion with heterogeneity in observable segments, rather than broad citywide expansion.

This section should also be read with appropriate discipline. Concentrated expansion is not the same as a welfare conclusion, and the city-year aggregate design does not by itself show whether total employment, worker surplus, or productivity rises at the city level. What it does show is a disciplined aggregate pattern: GGF exposure is associated with concentrated adjustment in observable labor-demand margins within exposed cities. That conclusion is fully consistent with the earlier evidence of strong treated-firm expansion, positive CSMAR top-five customer and supplier spillovers, and a procurement-based transmission channel, while the VC benchmark remains weak in the same spillover block. Taken together, the paper’s aggregate punchline is therefore about concentrated expansion with heterogeneity rather than uniform job creation.

8. Scope and Auxiliary Validation

We relegate tax-survey outcomes to the appendix rather than treating them as a second headline employment design. The tax match is narrower than the vacancy sample, and the matched employment series fails the long-window pre-trend test. Appendix Tables A3, A9, and A23 show that tax wage-per-worker remains the cleaner auxiliary outcome, while tax employment is best interpreted as directional evidence only. We therefore use the vacancy panel as the unique core sample for the main labor-demand results.

9. Conclusion

This paper evaluates the labor-market consequences of China’s Government Guidance Funds using firm-level vacancy data as the core empirical sample and using narrower administrative datasets only as auxiliary checks. Exploiting staggered investment timing, we document clear direct effects on treated firms and a broader pattern of concentrated expansion with heterogeneity across the local economy.

Four findings stand out. First, in the headline direct specification, GGF investment raises log hiring by 0.290 (SE 0.080) and log postings by 0.265 (SE 0.061), with selective wage/composition responses (0.105 on max wage, 0.091 on min wage, 0.053 on high-education share, and near-zero tech-post intensity at -0.007). Second, CSMAR top-five customer and supplier evidence points to observed-network spillovers. Customer coefficients are 0.330 (SE 0.173) for log hiring and 0.213 (SE 0.129) for log postings; supplier quantity estimates are also positive but less precise. The same-city $\times$ same-industry and IO-based BL/FL exercises do not support a broad positive spillover interpretation. Third, government procurement rises strongly along GGF-linked own-firm and customer-link paths (0.314 and 0.993 for log cumulative amount; 0.078 and 0.259 for log cumulative contracts), supporting a demand-channel interpretation for spillover transmission. Fourth, the refreshed GE table is consistent with concentrated expansion in observable margins but not with broad citywide growth: quantity effects are 0.091 (SE 0.108) in recruitment aggregates and 0.190 (SE 0.139) in listed-employee aggregates, while structure-share effects are -0.007 (SE 0.018) and -0.039 (SE 0.017).

The private-VC benchmark is informative but secondary. In the spillover comparison, private-VC on-impact estimates are small and statistically insignificant on quantity outcomes (event year 0: -0.011 for log hiring and -0.006 for log postings), so the benchmark does not overturn the GGF-centered interpretation.

These results highlight a key implication for policy evaluation: industrial policy can expand labor demand at targeted firms while simultaneously shifting that demand across firms and industries inside the local economy. Ignoring those concentrated-expansion margins risks overstating broad local job creation. The findings also demonstrate the value of high-

frequency vacancy data for monitoring labor-market responses to policy in near real time.

Limitations remain. The tax sample covers only a subset of firms and years (2007–2020), and the matched subsample is positively selected toward larger, more technology-intensive investees (Table A8); in addition, the matched employment series violates pre-trends. Vacancy data capture intended hiring rather than realized worker-level matches. Future work should link job-posting events to matched employer–employee administrative data to trace posting-to-employment conversion rates and wage-bill incidence at the worker level, and to quantify how GGF-induced skill upgrading propagates through occupation networks and supply chains. A separate line of work should study post-investment capital-market outcomes—IPO, acquisition, and exit—which are informative but conceptually distinct from the labor-demand responses that are the focus of this paper.

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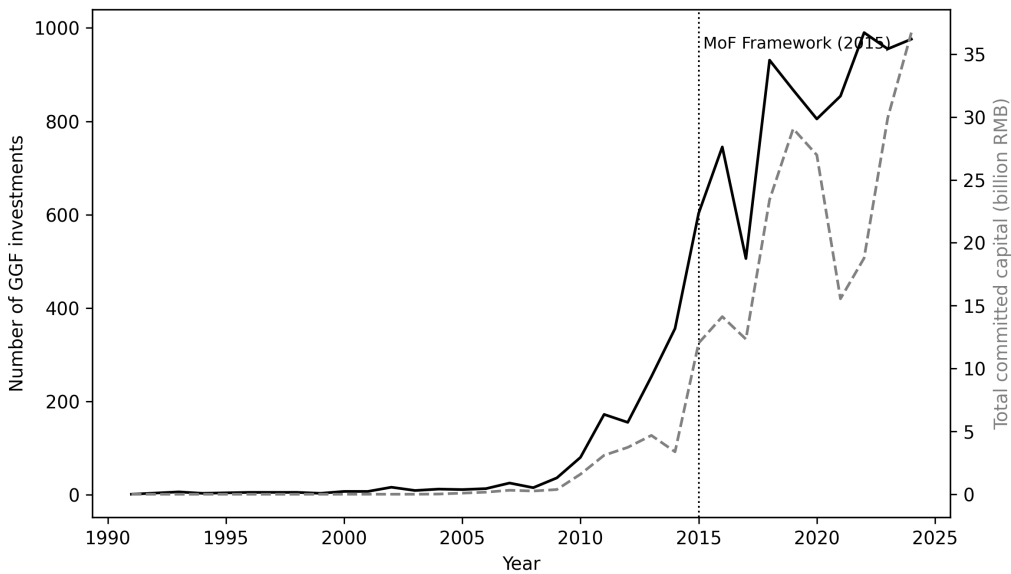
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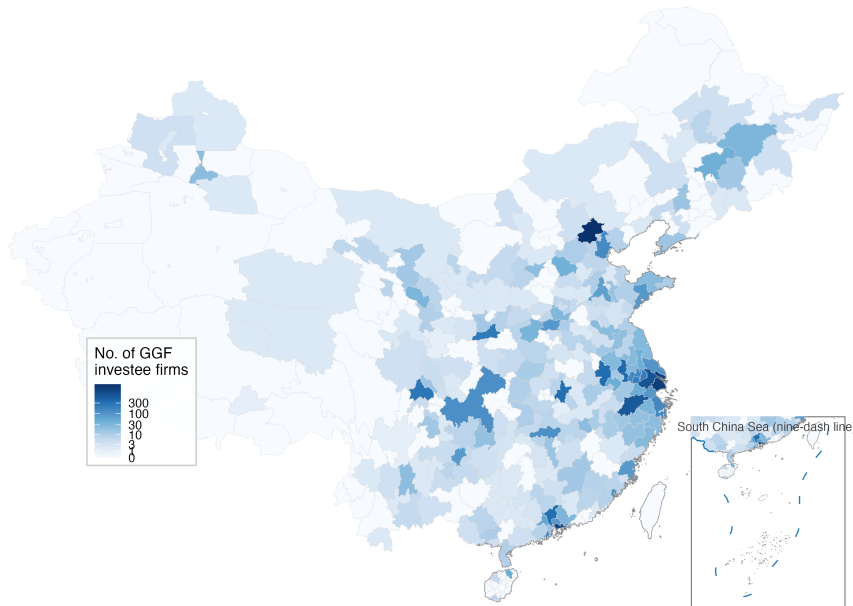
Figures and Tables



Source: Zero2IPO Database

(a) Trends in GGF and private VC investment activity

City-Level Distribution of GGF Investees
Mapped with 2024 administrative boundaries

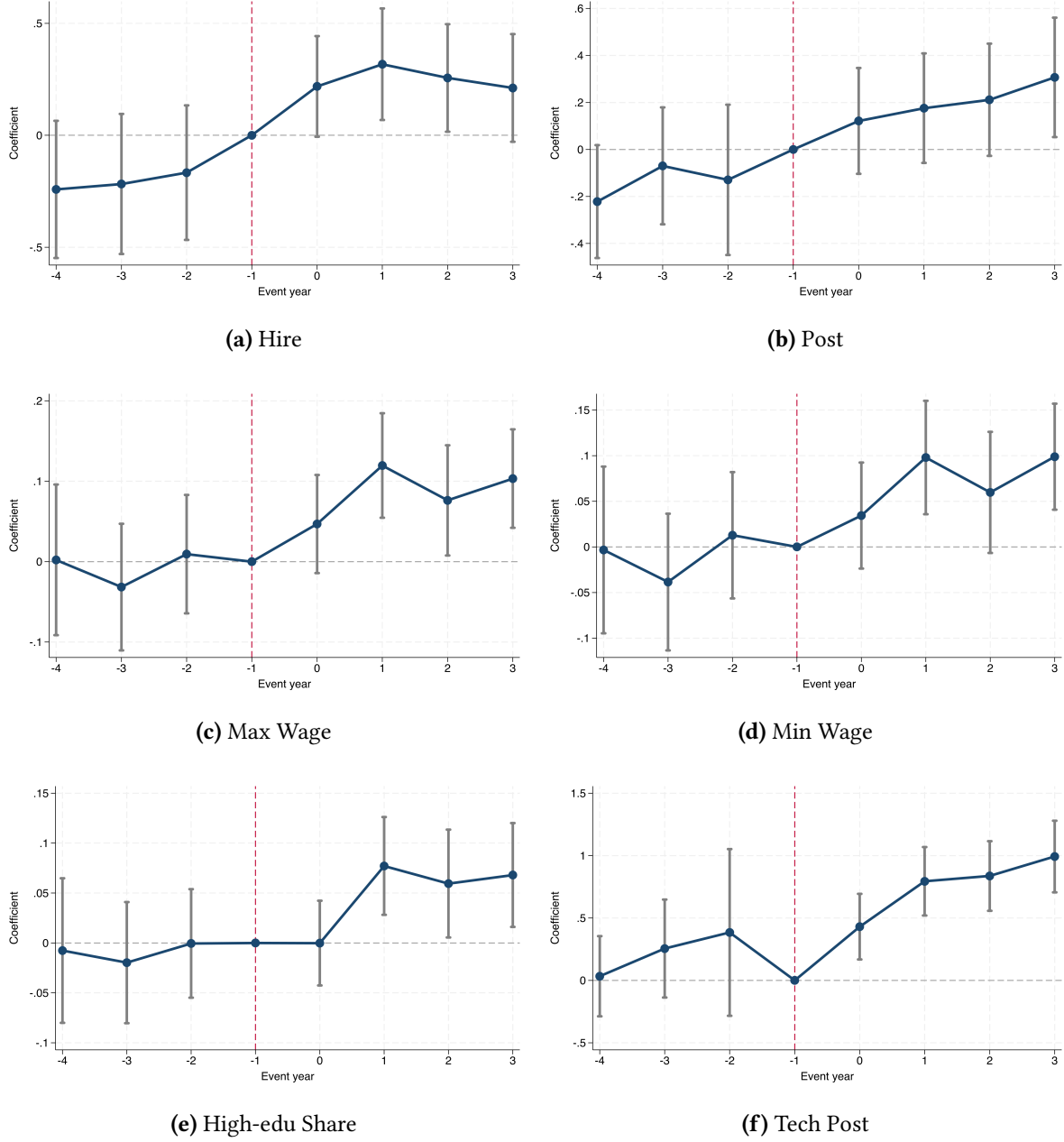


(b) Geographic distribution of GGF activity in China

Figure 1: Descriptive Trends of GGF and Private VC Investment Activity

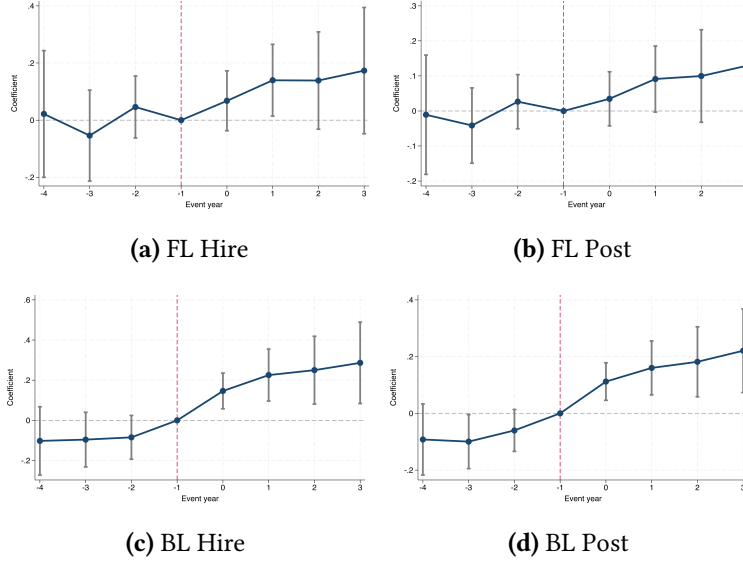
Notes: Panel (a) reports investment trends. Panel (b) maps the city-level distribution of GGF activity in China.

Figure 2: GGF Direct Event Study



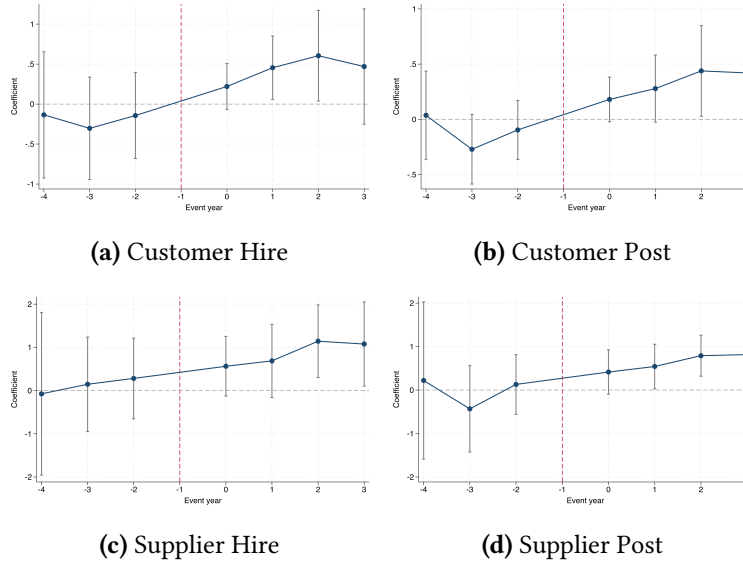
Notes: Panel estimates are event-study coefficients from firm-level staggered DiD regressions on the v6 recruitment panel. Period $k = -1$ is omitted. The event window is harmonized to $k \in [-4, 3]$ with endpoint binning. Regressions follow the baseline direct-effect specification used in the main text with firm-level treatment timing, include fixed effects as specified for each outcome in the locked Figure 2 setup, and report clustered standard errors at the firm level. Vertical bars denote 90% confidence intervals.

Figure 3: BL / FL Spillover Event Studies



Notes: Event-study coefficients for untreated IO-linked firms (FL/BL). Omitted period: $k = -1$. Window: $[-4, 3]$ with endpoint binning. Firm and province $\times$ year fixed effects; city-clustered standard errors; 90% confidence intervals.

Figure 4: CSMAR Top-Five Customer and Supplier Spillover Event Studies



Notes: Event-study coefficients for untreated CSMAR top-five customer and supplier counterparties. Omitted period: $k = -1$. Window: $[-4, 3]$ with endpoint binning. Role-specific locked timing/FE setup; city-clustered standard errors; 90% confidence intervals.

Figure 5: Government Procurement under GGF Supply-Chain Exposure



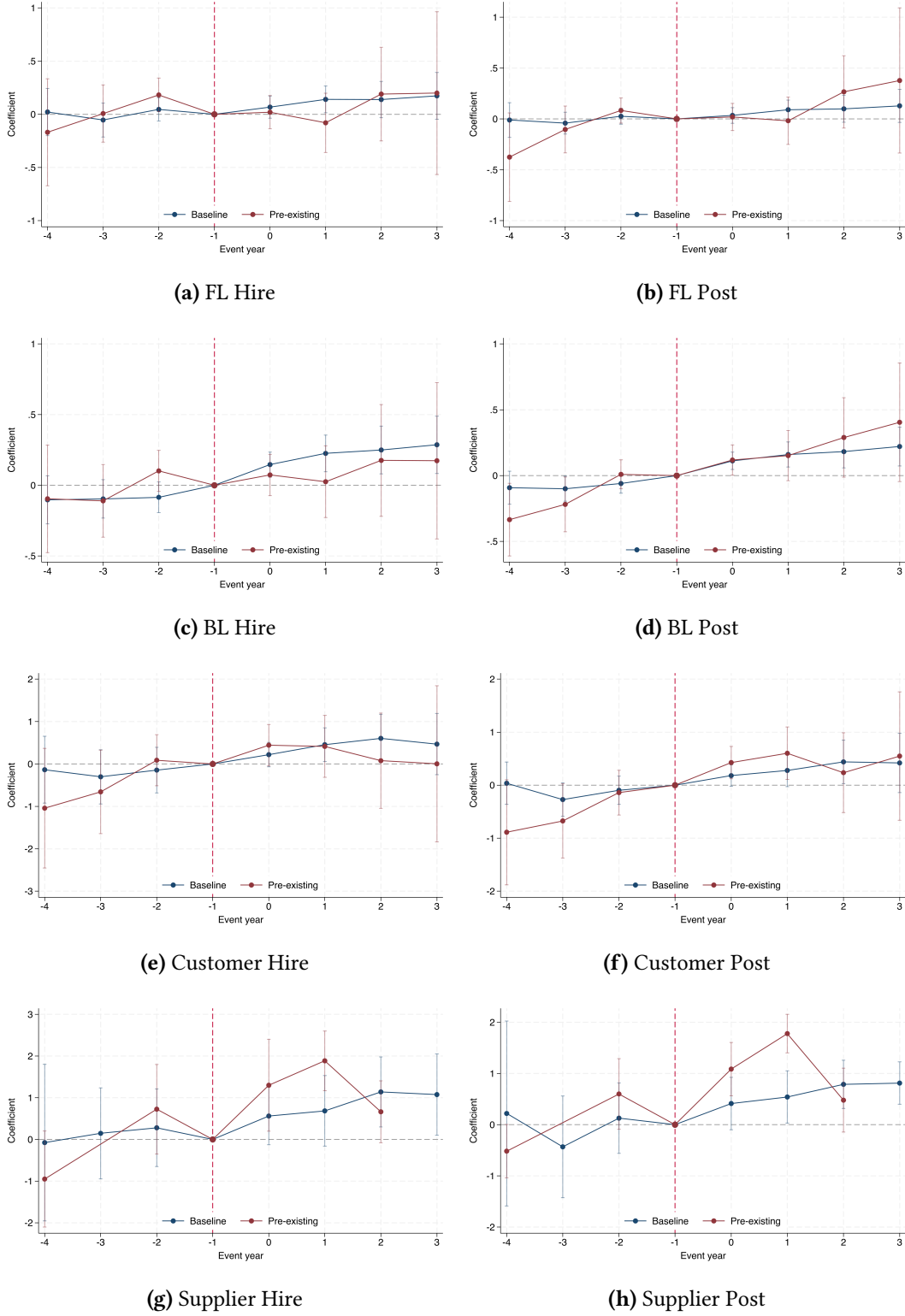
Notes: GGF mechanism event-study coefficients for cumulative procurement contracts and amounts. Omitted period: $k = -1$. Window: $[-4, 3]$. City-clustered standard errors; 90% confidence intervals.

Figure 6: Government Procurement under Private VC Supply-Chain Exposure



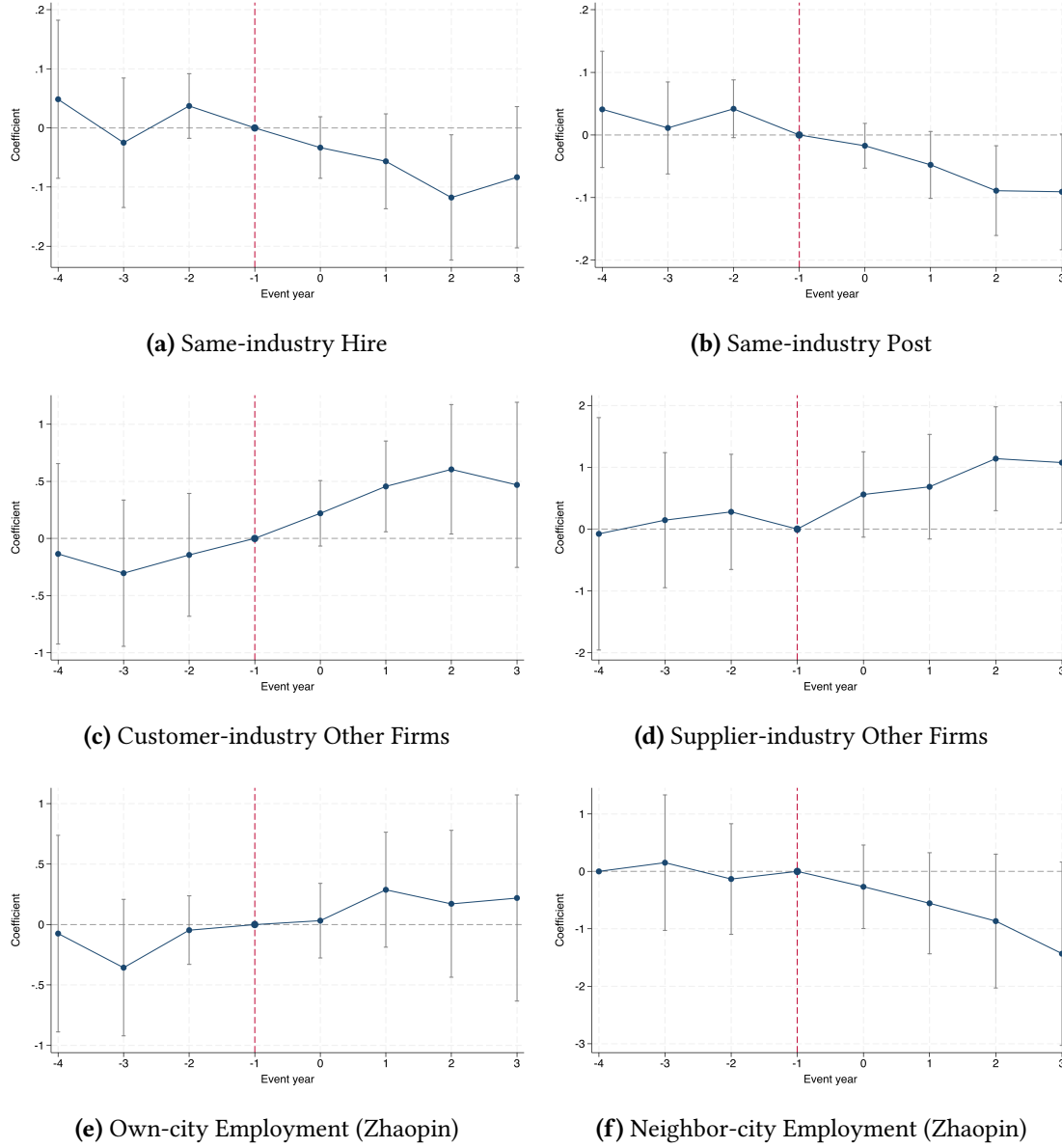
Notes: Private-VC benchmark event-study coefficients for the same procurement outcomes as Figure 5. Omitted period: $k = -1$. Window: $[-4, 3]$. City-clustered standard errors; 90% confidence intervals.

Figure 7: Pre-existing Spillover Event Studies (Figure 3/4 Matched Samples)



Notes: Each panel re-estimates the corresponding Figure 3/4 object on pre-existing samples only (firms observed at event time $k = -1$ within each design). Omitted period: $k = -1$. Event window: $[-4, 3]$ with endpoint binning. Regressions inherit the locked baseline timing and fixed-effects settings for each channel; standard errors are clustered at the city level. Vertical bars denote 90% confidence intervals.

Figure 8: GE Decomposition



Notes: Event-study decomposition with omitted period $k = -1$ and common plotting window $[-4, 3]$. Panels (a)–(b) report same-industry competition patterns. Panels (c)–(d) report responses for other firms in customer- and supplier-related industries. Panels (e)–(f) report own-city and neighboring-city employment responses using city-level Zhaopin aggregate hiring outcomes (not total employment-yearbook series). Vertical bars denote 90% confidence intervals.

Table 1: Sample Construction and Baseline Balance

Variable	Full Sample		GGF-Invested ($t = -1$)		Never-Treated		Diff (T-C)	p -value
	Mean	SD	Mean	SD	Mean	SD	Coef.	p -val
ln(Hiring)	1.633	0.884	2.049	0.948	1.631	0.883	0.418	0.000
ln(Posts)	1.568	0.864	1.974	0.934	1.566	0.862	0.408	0.000
ln(MaxWage)	8.988	0.901	9.354	0.646	8.986	0.901	0.367	0.000
ln(MinWage)	8.526	0.876	8.895	0.619	8.524	0.876	0.371	0.000
High-Skill Share	0.232	0.351	0.541	0.356	0.231	0.351	0.310	0.000
Tech-Posting Share	0.140	0.278	0.402	0.338	0.139	0.278	0.263	0.000
N Firms	1,006,764		603		1,004,640			
N Firm-Years	1,764,759		603		1,757,920			

Notes: The table reports descriptive statistics for the vacancy panel (2016–2024).

The treated column is evaluated at event time $t = -1$ (pre-treatment) to avoid post-treatment contamination.

Diff (T-C) is the treated-minus-never-treated mean difference; p -values are from large-sample tests.

Notes: The table reports baseline means and standard deviations for the matched vacancy sample used in the main analysis and serves as the main-text sample-construction and balance table. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 2: Direct ATEs of GGF

	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) ln(Min Wage)	(5) High-edu share	(6) Tech-post intensity
Treatment post	0.290*** (0.080)	0.265*** (0.061)	0.105* (0.063)	0.091 (0.060)	0.053** (0.022)	-0.007 (0.017)
Observations	184,318	184,318	171,566	171,445	184,318	184,318

Notes: This table reports GVC direct average treatment effects using the exact same treated/control sample, outcome definitions, and fixed-effects structure as Figure 2.

The sample uses GVC-backed firms as treated units and never-VC firms as controls, with treatment timing defined by first GVC investment under the manual-150 fund classification update.

All columns include firm and province $\times$ year fixed effects, cluster standard errors at the firm level, and apply the same event-window restriction used in Figure 2 (treated observations enter from event time -4 onward).

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: ATE estimates use the same treated/control sample, timing definition, outcomes, and firm plus province $\times$ year fixed-effects structure as Figure 2, with firm-clustered standard errors. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: CSMAR Top-Five Customer and Supplier Spillover ATEs

	ln(Hiring)	ln(Posts)	ln(Max Wage)	ln(Min Wage)
Panel A. CSMAR top-five customer spillover				
Post	0.330*	0.213	0.108	0.120*
	(0.173)	(0.129)	(0.075)	(0.069)
Observations	827	827	606	606
Panel B. CSMAR top-five supplier spillover				
Post	0.319	0.317	-0.084	-0.110
	(0.371)	(0.255)	(0.101)	(0.103)
Observations	130	130	123	123

Notes: Estimates are from the CSMAR top-five customer and supplier counterparty spillover design with untreated counterparties only. Standard errors are reported in parentheses.

Customer and supplier estimates are both used as observed-network spillover evidence. Customer estimates are more precise; supplier estimates are positive but less precise because the matched supplier sample is smaller.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Post-treatment ATEs for untreated CSMAR top-five customer and supplier counterparties linked to GGF-backed firms. Customer and supplier estimates are both reported as observed-network spillover evidence; customer estimates are more precise, while supplier estimates are positive but less precise due to the smaller matched sample. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Spillover ATEs with Taxonomy Structure

Panel A. Same-city $\times$ same-industry spillover ATE						
	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) ln(Min Wage)	(5) High-edu share	(6) Tech-post outcome
Spillover post	-0.029 (0.025)	-0.007 (0.014)	-0.001 (0.006)	0.004 (0.006)	-0.002 (0.003)	0.001 (0.003)
Observations	151,469	151,469	217,241	217,057	229,674	229,674
Fixed effects	Firm + Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year
Cluster	City	City	City	City	City	City
Panel B. BL spillover ATE						
	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) ln(Min Wage)	(5) High-edu share	(6) Tech-post outcome
Spillover post	-0.231*** (0.083)	-0.203*** (0.077)	-0.004 (0.021)	0.024 (0.022)	0.004 (0.010)	-0.015* (0.009)
Observations	222,455	222,455	262,557	215,890	267,089	267,089
Fixed effects	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year
Cluster	City	City	City	City	City	City
Panel C. FL spillover ATE						
	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) ln(Min Wage)	(5) High-edu share	(6) Tech-post outcome
Spillover post	-0.183*** (0.047)	-0.176*** (0.048)	0.005 (0.020)	0.009 (0.020)	-0.003 (0.014)	-0.006 (0.005)
Observations	264,246	264,246	259,355	259,302	219,003	263,724
Fixed effects	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year
Cluster	City	City	City	City	City	City
Panel D. Unrelated spillover ATE						
	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) ln(Min Wage)	(5) High-edu share	(6) Tech-post outcome
Spillover post	-0.033 (0.066)	-0.046 (0.058)	0.016 (0.015)	0.026 (0.018)	0.030** (0.012)	0.001 (0.008)
Observations	84,383	84,383	100,408	82,578	101,144	101,144
Fixed effects	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year	Firm + Province $\times$ Year
Cluster	City	City	City	City	City	City

Notes: Panel A matches the frozen Figure 3 spillover object exactly. Panels B–D use untreated firms only, GVC-only spillover timing, and the Figure 4 taxonomy groups defined inside the spillover sample: BL firms have positive backward linkage to treated sectors, FL firms have positive forward linkage, and unrelated firms have neither BL nor FL linkage.

Panel A retains its current live specification: same-city $\times$ same-industry exposure, event-time support $[-4, 3]$, city-clustered standard errors, and the outcome-specific fixed effects recorded in the current Table 3 metadata.

Panels B and D apply the frozen selected-spec family from the BL / unrelated event-study search. Panel C applies the frozen FL-only selected-spec family. Across Panels B–D, the table uses the same subgroup definitions, outcome transformations, fixed-effects choices, winsorization rules, and sample filters that feed the corresponding Figure 4 taxonomy panels; standard errors are clustered at the city level.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Panel A matches the frozen Figure 3 spillover object exactly. Panels B–D complete the BL / FL / unrelated taxonomy extension using untreated-firm spillover subsamples and the subgroup-specific Figure 4 spec family.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Heterogeneity / Composition Change

	(1)	(2)
	Composition share	Quantity anchor
Panel A. Recruitment Platform (v6)		
High-education posting share	0.008	
	(0.012)	
Technical posting share	-0.021	
	(0.010)	
Observations	31,391	
Firms	14,670	
Panel B. Listed Employee Structure		
BA+ share in employees	0.004	
	(0.010)	
ln(employee count)		0.168
		(0.048)
Observations	23,756	23,756
Firms	3,239	3,239

Notes: Coefficients are post-treatment DID estimates with firm and year fixed effects, clustered at the firm level.

Panel A uses the latest v6 recruitment firm-year panel. Panel B uses the listed-company employee panel from the established 2g matching chain.

The table is intended to update the composition narrative toward *concentrated expansion and structure change*, rather than citywide employment totals.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table occupies the main-text distributional / composition slot in the fixed architecture. Panel A asks whether the spillover quantity results survive on incumbent firms already present before exposure. Panel B asks which incumbent firms bear the incidence more strongly using a compact pre-exposure size split. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Mechanism: Government Procurement under GGF and Private VC Exposure

	ln(Cumulative procurement amount)	ln(Cumulative procurement contracts)
GGF own-firm exposure	0.314*** (0.052)	0.078*** (0.016)
GGF customer-link exposure	0.993*** (0.204)	0.259*** (0.062)
Private VC own-firm exposure	0.222*** (0.038)	0.049*** (0.010)

Notes: Post-treatment ATE summary from the procurement mechanism design. The table reports effect sizes for GGF own-firm, GGF customer-link, and private-VC own-firm exposures.

Standard errors are in parentheses. This table corresponds to Figure 5.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: The table reports post-treatment procurement responses for GGF own-firm exposure, GGF customer-link exposure, and private-VC own-firm exposure. It is designed to match Figure 5 and quantify channel strength by exposure type. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: GGF vs Private VC Benchmark

	ln(Hiring)	ln(Posts)
GGF spillover benchmark (event year 0)	-0.033 (0.032)	-0.017 (0.022)
Private VC spillover benchmark (event year 0)	-0.011 (0.020)	-0.006 (0.016)

Notes: The benchmark reports on-impact (event year 0) spillover coefficients from the matched GGF and private-VC spillover event-study comparison sample.

Both rows are statistically insignificant on quantity outcomes, so the benchmark is interpreted as a calibration object rather than an alternative headline design.

Standard errors are in parentheses. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Notes: This compact benchmark table keeps GGF as the main object and uses private VC as the external benchmark. The main-text comparison stays on the spillover-benchmark margin only; comparability diagnostics remain appendix support rather than part of the main benchmark block. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 8: GE / Aggregate Concentrated Expansion Headline Table

	Recruitment aggregate	Listed-employee aggregate
Post effect on quantity	0.091 (0.108)	0.190 (0.139)
Post effect on structure share	-0.007 (0.018)	-0.039 (0.017)
Avg observations per spec	2,360	2,360
Avg cities per spec	318	318
City FE	Yes	Yes
Year FE	Yes	Yes

Notes: This refreshed GE table is built from city-year aggregates available in two reproducible chains: (i) v6 recruitment postings and (ii) listed-company employee structure. Treatment timing is the first listed-GVC exposure year at the city level. Coefficients are post-treatment DID estimates with city and year fixed effects, clustered at city level. The object is interpreted as *concentrated expansion / structure change* in observable labor-demand margins, not citywide total employment.

Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Notes: This headline GE table matches Figure 8 and keeps the main-text aggregate object focused on concentrated expansion with heterogeneity across city-industry margins rather than diffuse citywide growth. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix

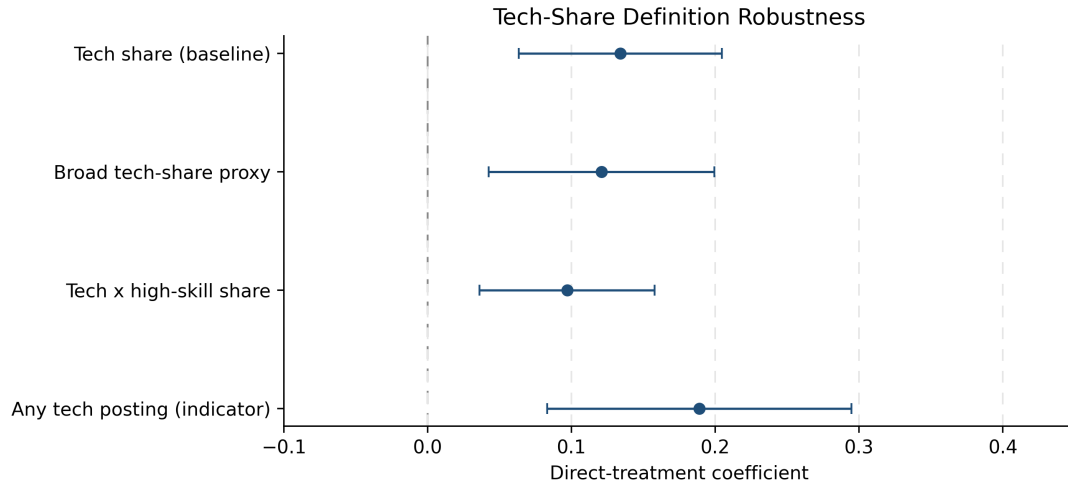


Figure A1: Tech Definition Robustness

Notes: Direct-effect coefficient estimates under alternative technical-posting definitions, redrawn in the same coefficient-dot-whisker format as Figures A2 and A3. The figure is intended as a compact robustness summary rather than a separate event-study panel.

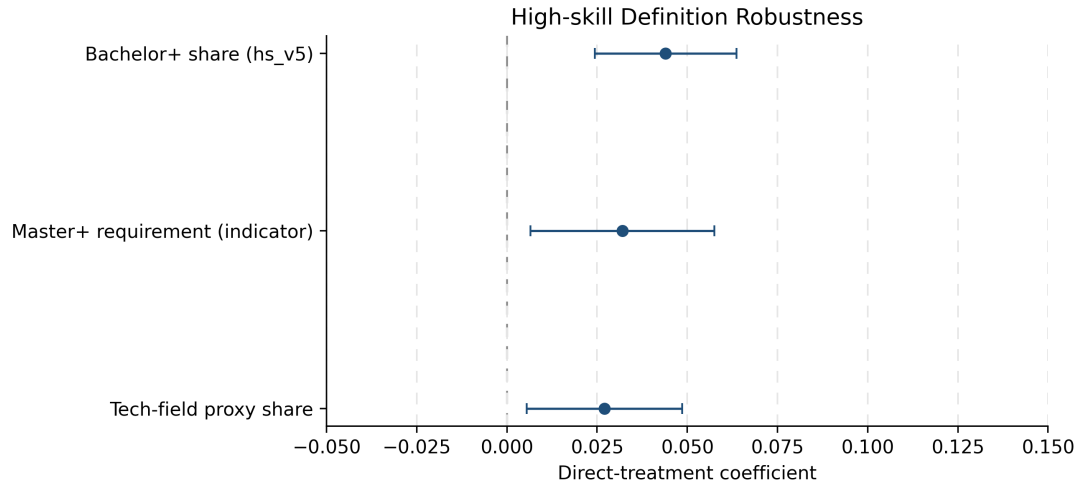


Figure A2: High-Skill Definition Robustness

Notes: Event-study coefficients for high-education vacancy share under alternative education-threshold definitions. Baseline specification as in Figure 2.

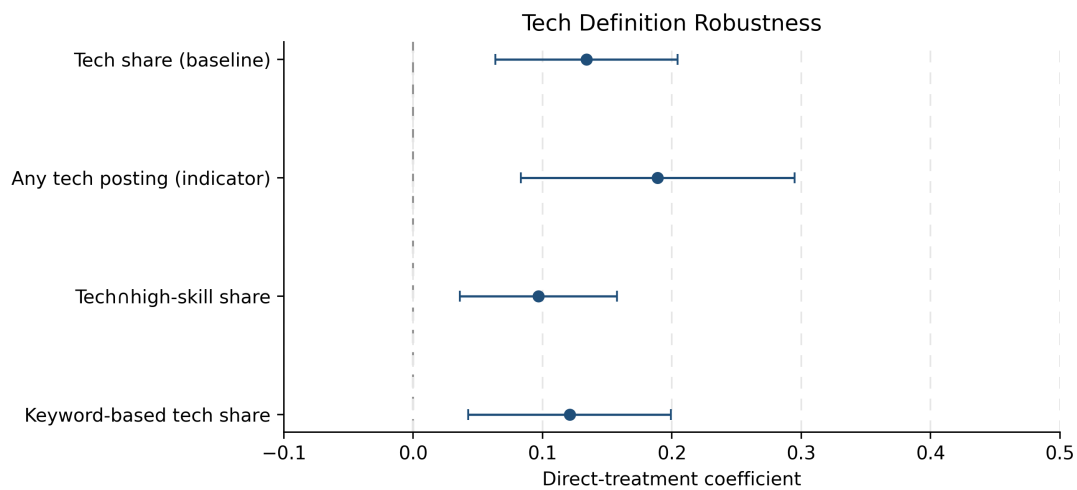


Figure A3: Tech Definition Robustness

Notes: Robustness of technical-posting definitions to alternative occupation classification schemes. Baseline specification as in Figure 2.

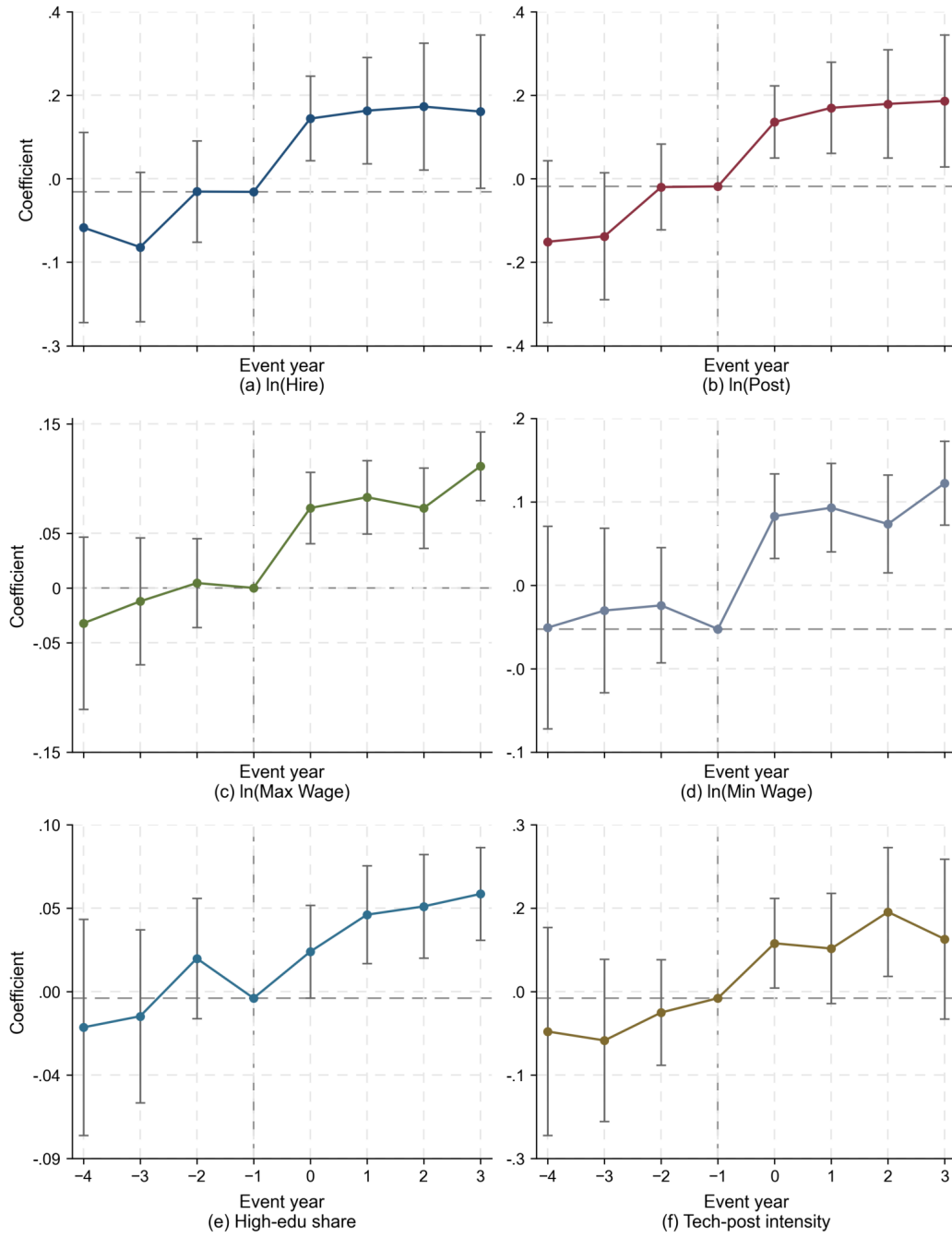


Figure A4: Sun-Abraham Event-Study Dynamics (Direct Effects)

Notes: Sun-Abraham (2021) heterogeneity-robust event-study estimates for direct effects. The omitted $t = -1$ period is normalized to zero, matching the presentation of the main event-study figures. This figure is meant to check robustness to staggered-adoption weighting, not to create a stronger visual pattern than the baseline event study.

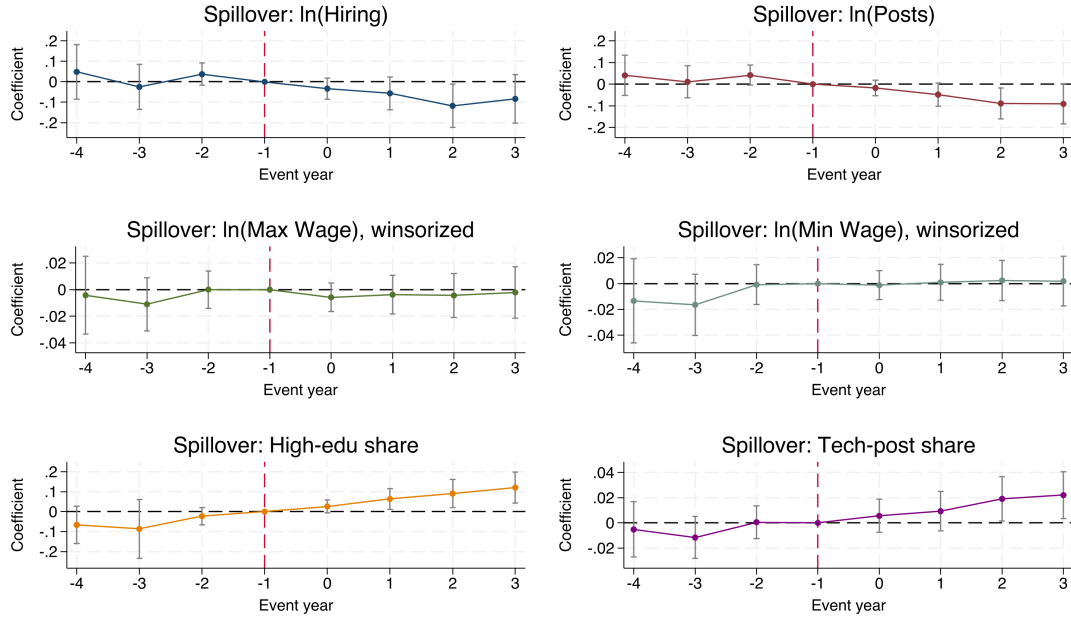


Figure A5: Same-City $\times$ Same-Industry Spillover Event Study (Appendix)

Notes: Event-study coefficients for the same-city $\times$ same-industry spillover object. The sample is restricted to untreated firms only, with GVC exposure defined by the first year in which any GVC-backed firm appears in the same local city-industry cell. This figure is kept in the appendix as a supporting robustness visual; the main text now emphasizes CSMAR top-five customer and supplier spillovers and IO (BL/FL) incidence as the core spillover block.

Notes on HonestDiD. HonestDiD figures should be read as sensitivity analyses for deviations from exact parallel trends. They are not alternative treatment-effect estimators; instead, they show how quickly the identified set widens as one allows progressively larger violations of the standard event-study assumptions.

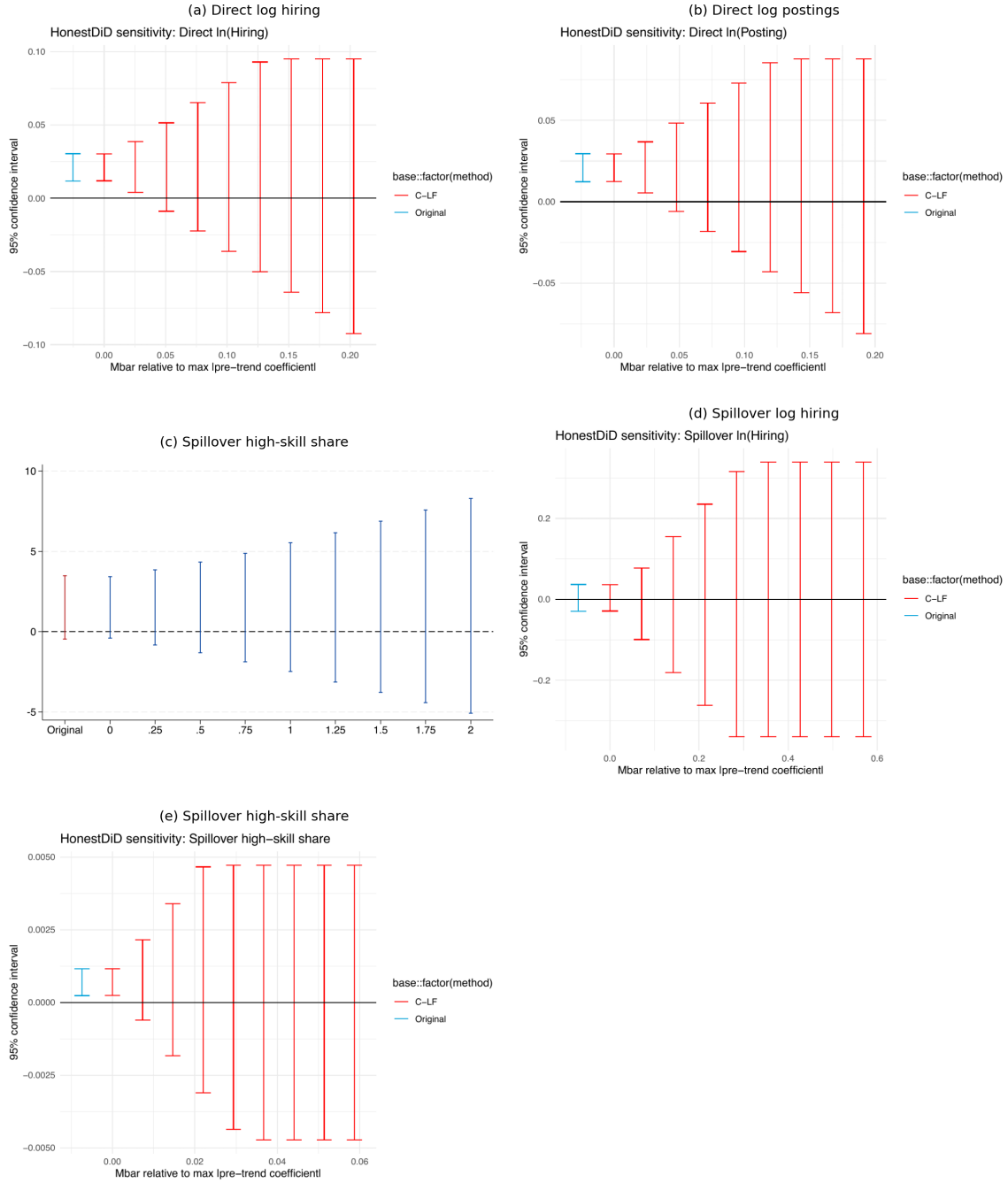


Figure A6: HonestDiD Sensitivity Bounds for Direct and Spillover Outcomes

Notes: Combined HonestDiD sensitivity bounds (Rambachan and Roth, 2023). Panel (a) reports the direct high-skill-share outcome; panels (b) and (c) report direct log hiring and log postings; panels (d) and (e) report spillover log hiring and spillover high-skill share. HonestDiD is a sensitivity analysis for deviations from exact parallel trends, not a new treatment-effect estimator. The spillover quantity interval includes zero already at $\bar{M} = 0$, while the direct vacancy outcomes and the spillover skill-composition outcome remain more stable.

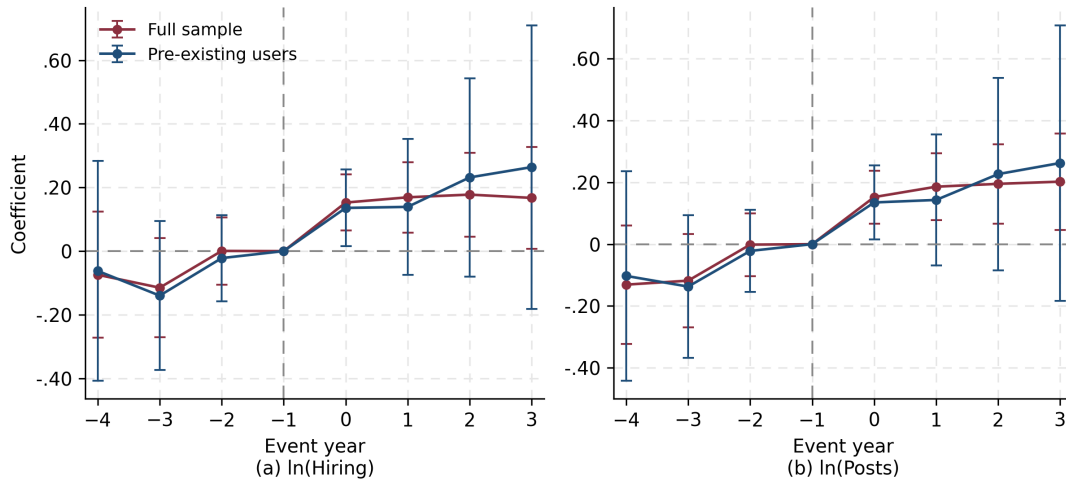


Figure A7: Platform-Margin Check: Full Sample versus Pre-Existing Platform Users

Notes: Event-study coefficients for direct vacancy outcomes under the baseline direct-effect specification. The red series uses the full treated sample, while the blue series restricts treated firms to those already observed on the recruitment platform before first GGF investment. The persistence of a post-treatment break in the pre-existing-user subsample weakens a pure platform-adoption interpretation of the direct vacancy effects.

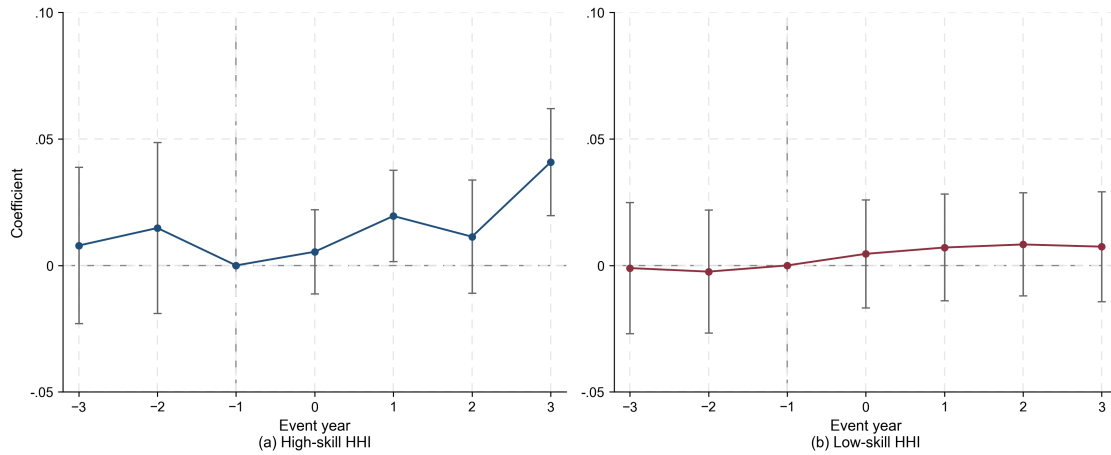


Figure A8: HHI / Concentration Dynamics After GGF Exposure

Notes: Event-study evidence for concentration dynamics in exposed markets. We treat this exercise as descriptive evidence on recruitment concentration within exposed markets rather than as a stand-alone welfare test of labor-market power.

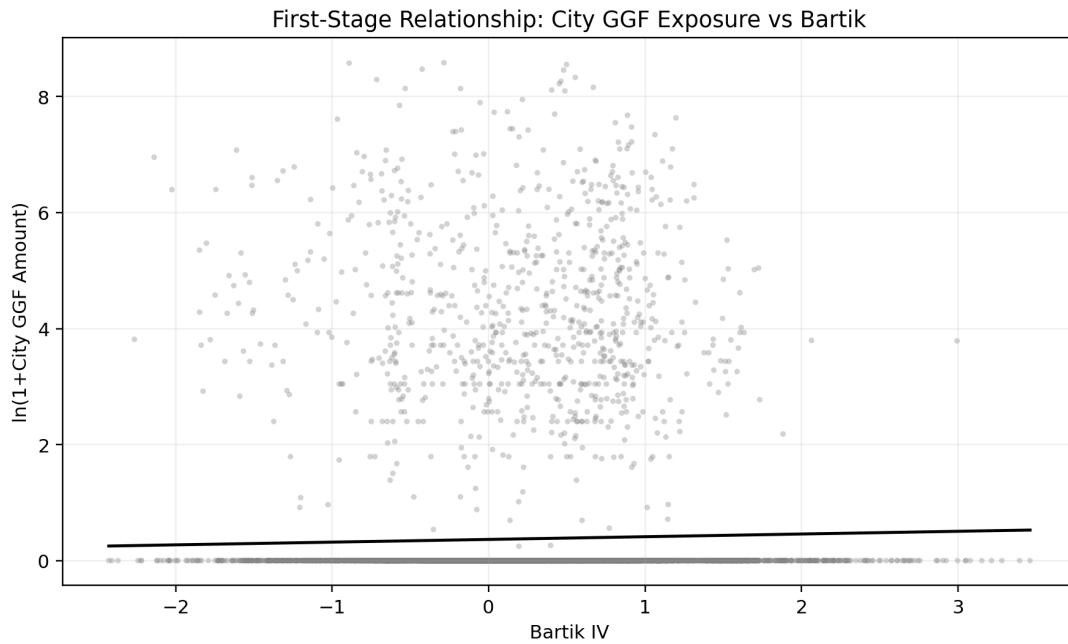


Figure A9: Bartik First-Stage Failure Diagnostic

Notes: City-level first-stage relationship between Bartik-style predicted exposure and realized GGF exposure. The relationship is weak and does not satisfy standard relevance thresholds, so the figure is kept only to document why the Bartik design is not used for identification.

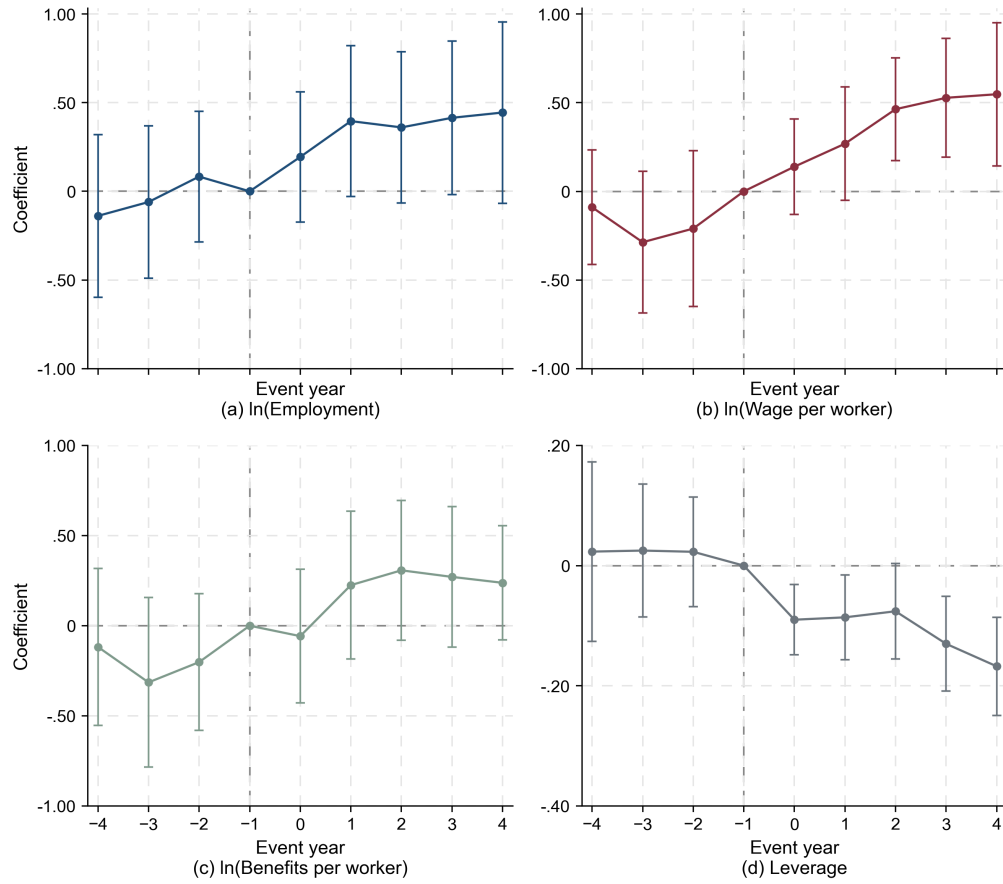
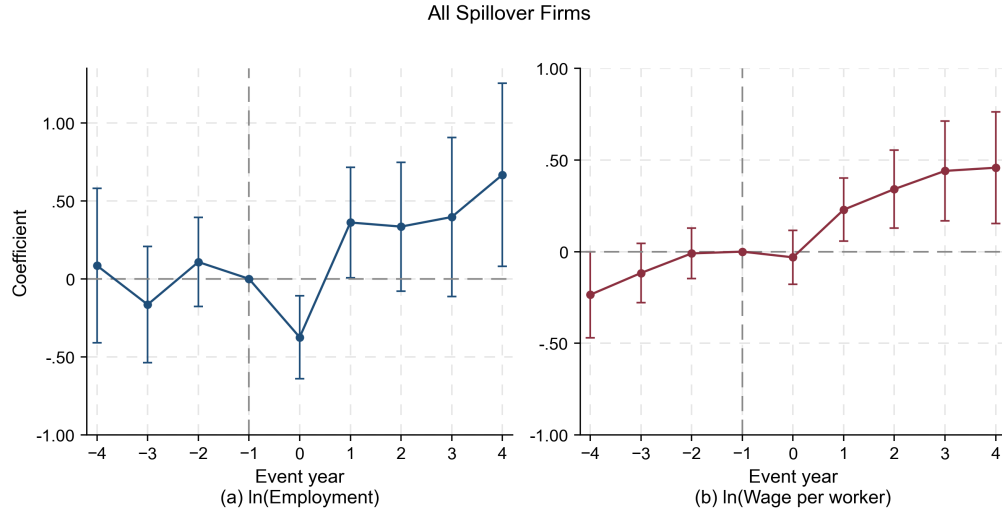


Figure A10: Auxiliary Tax-Survey Evidence: Event-Study Dynamics for Treated Firms

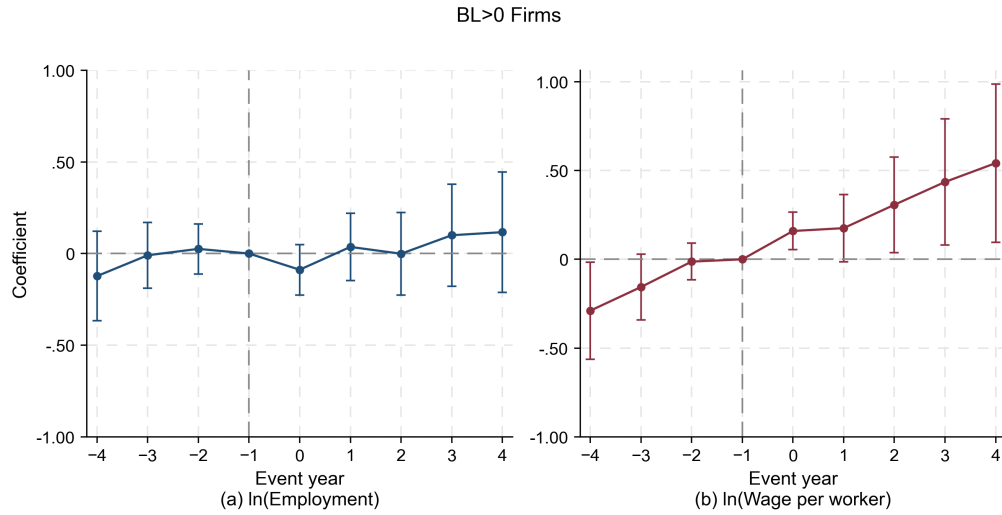
Notes: Tax-panel event-study estimates for treated firms. Period -1 is omitted. Regressions include firm and province-by-year fixed effects; standard errors are clustered at the firm level. These figures are retained as supportive evidence only because the matched tax employment series exhibits pre-trend failures.

Figure A11: Tax-Survey Spillover Diagnostic: All Non-Treated Firms in Exposed Cities



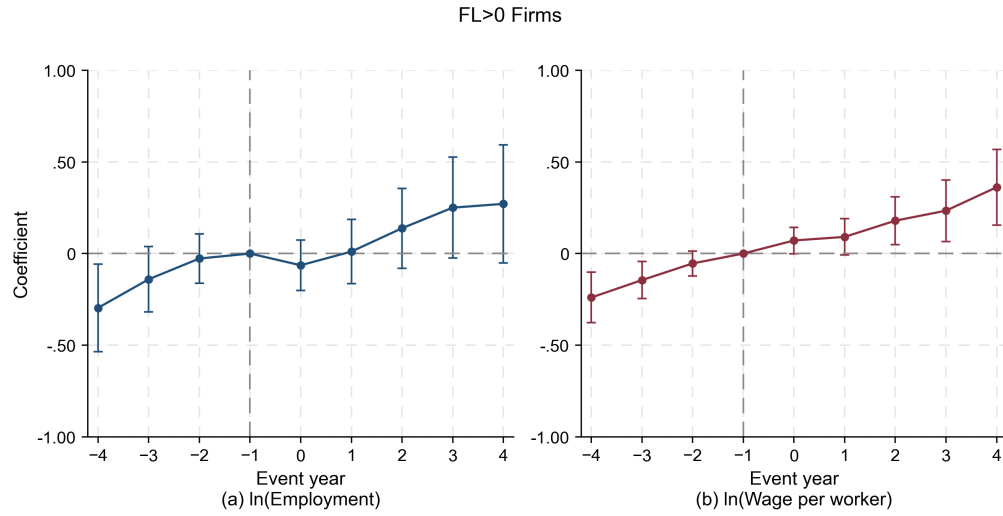
Notes: Event-study estimates using tax-survey outcomes for all never-treated firms in GGF-exposed cities. Outcomes are $\ln(\text{Employment})$ and $\ln(\text{Wage per worker})$. Event year runs from -4 to 4 with period -1 omitted. Regressions include firm and province-code $\times$ year fixed effects; standard errors use city-code clustering when feasible (with robust fallback for numerical singularity).

Figure A12: Tax-Survey Spillover Diagnostic: BL>0 Firms



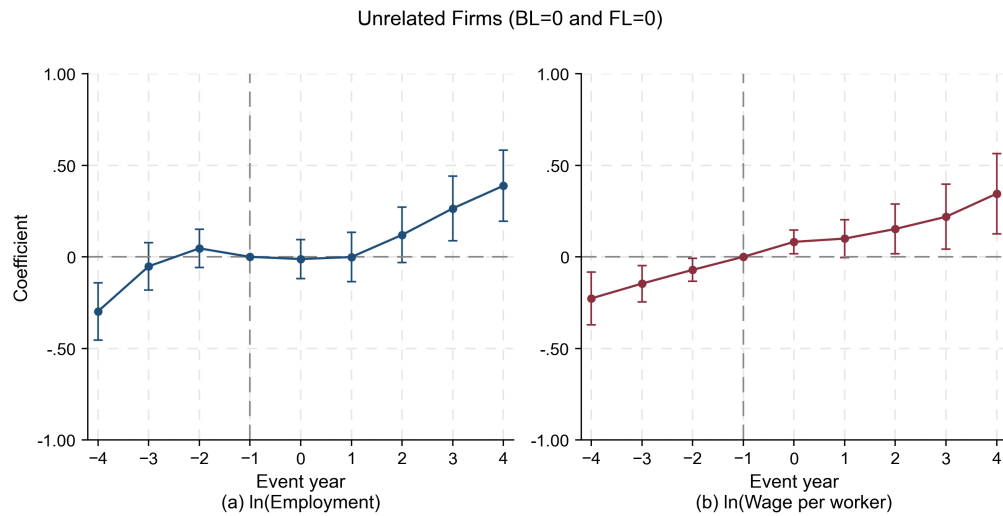
Notes: Event-study estimates for never-treated firms with positive backward linkage to treated sectors ($BL_i > 0$), where linkages are constructed from the 2020 National IO table and city treated-sector sets. Outcomes are $\ln(\text{Employment})$ and $\ln(\text{Wage per worker})$.

Figure A13: Tax-Survey Spillover Diagnostic: FL>0 Firms



Notes: Event-study estimates for never-treated firms with positive forward linkage to treated sectors ($FL_i > 0$), based on the same IO-linkage construction and regression specification as Figure A11. Outcomes are $\ln(\text{Employment})$ and $\ln(\text{Wage per worker})$.

Figure A14: Tax-Survey Spillover Diagnostic: Unrelated Firms ($BL=0$ and $FL=0$)



Notes: Event-study estimates for never-treated firms that are IO-unrelated to treated sectors ($BL_i = 0$ and $FL_i = 0$). Outcomes are $\ln(\text{Employment})$ and $\ln(\text{Wage per worker})$.

Table A1: Bartik First-Stage Failure Diagnostics

	(1) ln(1+City GGF Amount)	(2) City GGF Active (0/1)
bartik_iv	0.070 (0.063)	0.003 (0.014)
Observations	1753	1753

Standard errors in parentheses

City and year fixed effects included; standard errors clustered at city level.

Bartik instrument uses pre-period city industry shares and national GGF industry growth.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The instrument is a Bartik-style predicted city exposure, constructed as pre-period city industry shares interacted with national GGF growth by industry. Columns report first-stage relationships with city-level realized GGF exposure outcomes. Regressions include city and year fixed effects with city-clustered standard errors. Neither column satisfies conventional relevance thresholds, so we do not use this design for identification.

Table A2: Bartik Reduced-Form Correlations on Vacancy Outcomes

	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) High-skill posts
bartik_iv	0.089* (0.047)	0.088* (0.047)	-0.001 (0.006)	0.493* (0.295)
Observations	927907	927907	863030	927907

Standard errors in parentheses

Never-treated sample; firm and time fixed effects per selected spec; city-clustered SE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Reduced-form regressions of vacancy outcomes on the Bartik predictor in the never-treated sample. Regressions include firm and year fixed effects with city-clustered standard errors. Because the first stage is weak, these coefficients should be read as descriptive correlations rather than IV evidence.

Table A3: Tax-Survey Financial Outcomes (Appendix)

	(1) ln(Revenue)	(2) ln(Profit)	(3) ln(Assets)	(4) Leverage
GGF Investment	-0.050 (0.262)	-1.247 (1.168)	-0.662* (0.380)	-0.087* (0.050)
Observations	279	279	279	279
Firm FE	✓	✓	✓	✓
Province×Year FE	✓	✓	✓	✓

Notes: The table reports treatment effects on tax-survey financial outcomes for GGF-invested firms. Variable definitions follow the tax data dictionary; fixed effects and clustering follow the baseline tax specification. Firm and province×year fixed effects included. Standard errors clustered at the firm level in parentheses. Significance levels:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Tax-Survey Pre-trend Event-Study Tests (Six Outcomes)

	(1) ln(Revenue)	(2) ln(Profit)	(3) ln(Assets)	(4) Leverage	(5) ln(Employment)	(6) ln(Wage p.c.)
Event time: ≤ -3	-1.010*** (0.124)	-0.865 (0.611)	-0.751*** (0.093)	0.016 (0.026)	-0.391*** (0.091)	-0.098 (0.078)
Event time: -2	-0.564*** (0.150)	-0.703* (0.420)	-0.478*** (0.138)	0.010 (0.023)	-0.263*** (0.074)	-0.138 (0.084)
Event time: 0	0.182** (0.086)	-0.604** (0.301)	0.179*** (0.048)	-0.057*** (0.021)	0.143** (0.061)	0.103 (0.072)
Event time: $+1$	0.279** (0.116)	-1.410*** (0.433)	0.329*** (0.067)	-0.042** (0.020)	0.115 (0.076)	0.126 (0.093)
Event time: $+2$	0.401*** (0.113)	-1.941*** (0.519)	0.498*** (0.076)	-0.037 (0.023)	0.216** (0.100)	0.154 (0.100)
Event time: $\geq +3$	0.474*** (0.148)	-2.675*** (0.545)	0.504*** (0.092)	-0.056** (0.025)	0.263** (0.113)	0.095 (0.108)
Pre-trend F -test (p)	0.000	0.204	0.000	0.826	0.000	0.246
Observations	1499336	1500080	1500005	1478675	1483513	1477136
Firms	375018	375020	375017	370942	372279	370877

Standard errors in parentheses

Tax survey sample (2007–2020). Firm and province×year FE included.

Omitted category: event time -1 .

Pre-trend F -test: joint significance of event times ≤ -3 and -2 .

Columns (1)–(4): mechanism/descriptive outcomes (revenue and assets fail pre-trends, consistent with selection-on-dip; not used causally).

Columns (5)–(6): main causal outcomes (Table 4); employment rejects pre-trends while wage does not.

SE clustered at firm level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table reports event-time coefficients and joint pre-trend tests in the tax panel. Revenue, assets, and employment fail the pre-trend test, while wage per worker is comparatively stable pre-treatment; accordingly, the tax employment series is treated as directional rather than clean validation. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Auxiliary Tax-Survey Evidence: Direct Effects on Invested Firms

	(1) ln(Employment)	(2) ln(Wage p.c.)	(3) Amenity per worker	(4) Leverage
Treated $\times$ Post	0.465*** (0.097)	0.148*** (0.051)	0.144 (0.126)	-0.047** (0.023)
Observations	21794	17459	6144	6850

Standard errors in parentheses

Firm FE and province $\times$ year FE included.

SE clustered at firm level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Tax direct-effect estimates use the matched tax panel with firm fixed effects and province-by-year fixed effects, clustering standard errors at the firm level. These estimates are auxiliary because the matched employment series exhibits pre-trend violations. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Tax Spillover Diagnostics: Non-Invested Firms in GGF-Active Cities

	(1) ln(Employment)	(2) ln(Wage p.c.)
City GGF exposure (post)	0.022* (0.011)	0.022 (0.052)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Mean dep. var.	2.926	3.561
Observations	421827	419616

Notes: The sample is restricted to never-treated firms observed in GGF-exposed cities. Treated cities are those with first GGF investment in 2018 or 2019, and control cities have no GGF exposure in the sample window. Firm and year fixed effects are included. Standard errors are clustered at the city level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Tax-sample spillover estimates are based on never-treated firms in exposed cities with fixed effects aligned to the tax panel and clustered inference at the firm level. These estimates are best read as auxiliary because industry-level tax matching and spillover composition are both limited in the tax data. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Tax Spillover Validation (Alternative Full-Sample Definition)

	(1) ln(Employment)	(2) ln(Wage p.c.)	(3) Amenity per worker	(4) Leverage
Spillover (post)	0.094 (0.119)	-0.027 (0.025)	-0.085 (0.071)	-0.031** (0.013)
Observations	3975591	3026235	941790	1159058

Standard errors in parentheses

Firm FE and province-by-year FE included.

SE clustered at city level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the broader tax spillover specification using the strict full sample of never-treated firms in exposed cities. It is kept in the appendix as a robustness reference; the main text uses Table A6 as the headline tax-spillover specification.

Table A8: Tax-Survey Sample Representativeness: Matched vs. Unmatched Firms

Variable	Matched	Unmatched	Diff	t-stat	p-value	SMD
ln(Hiring)	1.906	1.622	0.285***	-62.755	0.0000	0.329
ln(Job Posts)	1.727	1.559	0.167***	-39.503	0.0000	0.202
ln(Max Wage)	8.850	8.990	-0.140***	48.732	0.0000	-0.193
ln(Min Wage)	8.402	8.527	-0.126***	39.410	0.0000	-0.172
High-skill Share	0.129	0.119	0.009***	-5.158	0.0000	0.028
Tech Share	0.177	0.139	0.037***	-24.621	0.0000	0.133

Notes: "Matched" indicates firms with tax-survey linkage (`has_tax_data=1`). "Diff" reports matched-minus-unmatched means.

SMD = standardized mean difference using pooled SD.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: The table compares pre-treatment characteristics of GGF-investee firms that can be matched to the tax-survey data (*Matched*) against those that cannot (*Unmatched*), using vacancy-sample outcomes measured at $t = -1$. Diff reports the matched-minus-unmatched mean difference; SMD is the standardized mean difference using pooled standard deviation. All differences are statistically significant, indicating that the tax-validated subsample is positively selected on hiring intensity and technology composition. Tax-survey estimates therefore represent a local average treatment effect for the matched (larger, more tech-intensive) subgroup rather than an average effect for all GGF investees.

Table A9: Coverage of Auxiliary Samples Relative to VC- and GVC-Backed Firms

Sample	Unique firms	Firms with any VC	Firms with GVC	Share of GVC firms covered
Recruitment panel	804,770	15,811	4,338	0.323
Tax panel	1,625,873	10,446	2,991	0.223
Listed parents	5,263	2,976	962	0.183

Notes: The table compares the three auxiliary firm-level samples that are most relevant for this paper: the cleaned recruitment panel, the matched tax panel, and the listed-parent registry sample. The final column reports the share of all GVC-backed firms in the enterprise-level VC master that appear in each auxiliary sample. The recruitment panel offers the broadest labor-demand coverage and is therefore the paper's unique core sample. Tax and listed-firm data are retained for supportive corroboration and heterogeneity only.

Table A10: PPML Robustness for Count Outcomes

	(1) Dir: Hiring	(2) Dir: Posts	(3) Dir: HS posts	(4) Dir: Tech posts	(5) Spill: Hiring	(6) Spill: Posts	(7) Spill: HS posts	(8) Spill: Tech posts
did_direct	0.183*** (0.068)	0.214*** (0.061)	0.221*** (0.070)	0.088 (0.077)				
spill_post					0.068 (0.082)	0.041 (0.086)	0.097* (0.055)	0.009 (0.049)
Observations	5836	5481	5337	5056	1112142	1111164	173950	139803

Standard errors in parentheses

PPML with high-dimensional FE. Direct: firm FE + province-year FE, clustered by firm.

Spillover: firm FE + year FE, clustered by city.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: PPML specifications re-estimate the main count outcomes without log-linearization. Fixed effects and clustering follow the baseline direct-effects design, and the table checks that the positive hiring and posting responses are not driven by the log transformation. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Platform-Margin Decomposition of Direct Vacancy Effects

	ln(Hiring)	ln(Posts)
Full sample	0.189*** (0.042)	0.195*** (0.041)
Pre-existing platform users	0.134** (0.053)	0.129** (0.052)
New adopters	0.243 (0.194)	0.289 (0.190)
Observations	1,764,485	1,764,485

Notes: The full-sample estimate captures the total vacancy response observed on the platform. The pre-existing-user estimate more cleanly isolates within-platform recruitment intensity. If post-treatment platform adoption partly reflects real expansion, the underlying labor-demand effect need not equal either estimate exactly, so we report both without treating either as mechanically correct.

Table A12: Direct-Effect Robustness Checks

	(1) Baseline	(2) City Cluster	(3) Excl. COVID	(4) PPML	(5) Drop top 5%
GGF Investment	0.198*** (0.045)	0.198*** (0.040)	0.202*** (0.052)	0.183*** (0.068)	0.168*** (0.056)
Observations	5,836	5,836	4,778	5,836	2,421
Firm FE	✓	✓	✓	✓	✓
Province × Year FE	✓	✓	✓	✓	✓

Notes: This table reports robustness checks for the direct-effects baseline with dependent variable $\ln(\text{Hiring} + 1)$. Columns vary clustering and sample restrictions as indicated in column headers. Unless otherwise noted, specifications include the baseline fixed effects. Standard errors are clustered as reported, and significance levels are $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A13: Vacancy-Sample Event-Study Joint Tests

Outcome	Pre-trend F-test p -value	Joint post-treatment p -value
<i>Direct effects</i>		
$\ln(\text{Hiring})$	0.199	0.003
$\ln(\text{Posts})$	0.157	0.006
$\ln(\text{Max Wage})$	0.657	0.053
$\ln(\text{Min Wage})$	0.964	0.125
High-skill share	0.110	0.020
Tech postings	0.274	0.044
<i>Spillover effects</i>		
$\ln(\text{Hiring})$	0.238	0.156
$\ln(\text{Posts})$	0.415	0.195
$\ln(\text{Max Wage})$	0.571	0.424
$\ln(\text{Min Wage})$	0.877	0.631
High-skill postings	0.396	0.021
Tech postings	0.770	0.019

Notes: The table reports joint significance tests from event-study specifications. “Pre-trend” tests jointly evaluate all leads ($k \leq -2$); “Post” tests jointly evaluate all lags ($k \geq 0$). Entries are computed from `Tables/v3/quick_es_pvals_all.csv`.

Notes: Joint tests are computed from the vacancy event studies used in the main text. Pre-treatment coefficients are jointly insignificant for the direct vacancy outcomes, supporting the identifying parallel-trends assumption in the main platform panel. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A14: GVC vs Private VC: Pre-Treatment Comparability of Treated Firms

	GVC-only	Private-VC-only	Diff	SMD
ln(Hiring) at $t = -1$	2.662 (1.108)	2.430 (1.169)	0.231	0.199
ln(Posts) at $t = -1$	2.175 (0.899)	1.986 (0.927)	0.189	0.204
ln(Max Wage) at $t = -1$	9.220 (0.416)	9.201 (0.668)	0.019	0.029
ln(Min Wage) at $t = -1$	8.785 (0.383)	8.758 (0.651)	0.027	0.043
High-edu share at $t = -1$	0.377 (0.355)	0.462 (0.359)	-0.085	-0.238
Tech-post share at $t = -1$	0.234 (0.233)	0.300 (0.313)	-0.066	-0.216
GVC-only firms	47			
Private-VC-only firms		386		

Notes: The table compares mutually exclusive GVC-only treated firms with mutually exclusive private-VC-only treated firms at the last pre-treatment observation ($t = -1$) in the vacancy panel.

Diff reports the GVC-only minus private-VC-only mean difference. SMD is the standardized mean difference using the pooled standard deviation. This object is intended as comparability support for the main-text benchmark table rather than as a separate selection model.

Significance levels for Diff: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Notes: This appendix table supports the main-text benchmark by comparing GVC-only and private-VC-only treated firms on key vacancy observables measured at $t = -1$. It is intended to show how similar the two treated groups look before treatment, not to claim random assignment across capital types.

Table A15: GVC vs Private VC: Within-Industry-Year Benchmark Support

	(1) ln(Hiring)	(2) ln(Posts)	(3) ln(Max Wage)	(4) ln(Min Wage)	(5) High-edu share
Baseline benchmark gap	-0.046 (0.171)	-0.044 (0.120)	-0.149 (0.093)	-0.123 (0.091)	0.024 (0.045)
Within-industry-year FE gap	-0.043 (0.171)	-0.038 (0.124)	-0.129 (0.094)	-0.103 (0.092)	0.015 (0.044)
Observations	185,975	185,975	173,102	172,977	185,975

Notes: The table reports the GVC-minus-private-VC benchmark gap under the baseline stacked benchmark specification and under a stronger specification that adds industry×year fixed effects to absorb common industry-time composition shifts.

Both specifications retain firm and province×year fixed effects, cluster at the firm level, use mutually exclusive GVC-only and private-VC-only treated firms, and compare them against the common never-VC control pool.

This object is designed to show whether the main benchmark comparison is mostly a sector-composition artifact. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table asks whether the compact Table 7 benchmark gap survives after absorbing industry×year shocks. It is an appendix support object for external-validity interpretation, not a replacement for the main benchmark table.

Table A16: Industry Distribution of GGF Investments

Rank	Industry category	Classification label	Investments	Share (%)
1	Information services	Information Transmission, Computer Services and Software Computer Services	965	10.2
2	Internet information services	Information Transmission, Computer Services and Software Telecommunications and Other Information Transmission Services Internet Information Services	857	9.1
3	Manufacturing (aggregate)	Manufacturing	773	8.2
4	Integrated circuits	Manufacturing Communication Equipment, Computers and Other Electronic Equipment Electronic Components Integrated Circuit Manufacturing	704	7.5
5	Medical instruments	Manufacturing Special Equipment Manufacturing Medical Instruments and Equipment	508	5.4
6	Instrumentation and office machinery	Manufacturing Instrumentation, Cultural and Office Machinery	458	4.9
7	Chemical products	Manufacturing Chemical Raw Materials and Chemical Products	429	4.5
8	Electronics equipment	Manufacturing Communication Equipment, Computers and Other Electronic Equipment	422	4.5
9	Pharmaceuticals	Manufacturing Pharmaceutical Manufacturing	407	4.3
10	Biopharmaceuticals	Manufacturing Pharmaceutical Manufacturing Biological and Biochemical Products	306	3.2
11	Software industry	Information Transmission, Computer Services and Software Software Industry	215	2.3
12	Computer manufacturing	Manufacturing Communication Equipment, Computers and Other Electronic Equipment Computer Manufacturing	200	2.1
13	Health services	Health, Social Security and Social Welfare Health Services	180	1.9
14	Electrical machinery	Manufacturing Electrical Machinery and Equipment	167	1.8
15	Other financial activities	Finance Other Financial Activities	163	1.7

Continued on next page

Table A16: Industry Distribution of GGF Investments (continued)

Rank	Industry category	Classification label	Investments	Share (%)
16	Automobile manufacturing	Manufacturing Transport Equipment Manufacturing Automobile Manufacturing	153	1.6
17	Environmental management	Water, Environment and Public Facility Management Environmental Management Environmental Governance	145	1.5
18	Power generation	Electricity, Gas and Water Production Power and Thermal Production Power Generation	142	1.5
19	Medical supplies	Manufacturing Pharmaceutical Manufacturing Sanitary Materials and Medical Supplies	130	1.4
20	Agriculture	Agriculture, Forestry, Animal Husbandry and Fishery Agriculture	104	1.1
21	Semiconductor discrete devices	Manufacturing Communication Equipment, Computers and Other Electronic Equipment Electronic Device Manufacturing Semiconductor Discrete Devices	98	1.0
22	Basic chemical materials	Manufacturing Chemical Raw Materials and Chemical Products Basic Chemical Materials	98	1.0
23	Culture, sports, and entertainment	Culture, Sports and Entertainment	96	1.0
24	Electronic components	Manufacturing Communication Equipment, Computers and Other Electronic Equipment Electronic Components Electronic Parts and Assemblies	83	0.9
25	Application software services	Information Transmission, Computer Services and Software Software Industry Public Software Services Application Software Services	79	0.8
26	Unclassified	–	62	0.7
27	Optoelectronic devices	Manufacturing Communication Equipment, Computers and Other Electronic Equipment Electronic Device Manufacturing Optoelectronic Devices	61	0.6
28	Retail trade	Wholesale and Retail Trade Retail Trade	59	0.6
29	Construction engineering	Construction Building and Civil Engineering	58	0.6
30	Film and broadcasting	Culture, Sports and Entertainment Broadcasting, TV, Film and Audio-Visual	58	0.6
31	Education	Education	54	0.6

Continued on next page

Table A16: Industry Distribution of GGF Investments (continued)

Rank	Industry category	Classification label	Investments	Share (%)
32	Basic software services	Information Transmission, Computer Services and Software Software Industry Public Software Services Basic Software Services	51	0.5
33	Construction (aggregate)	Construction	51	0.5
34	Arts and culture	Culture, Sports and Entertainment Arts and Culture	46	0.5
35	Food manufacturing	Manufacturing Food Manufacturing	41	0.4
36	Agricultural food processing	Manufacturing Agricultural and Sideline Food Processing	39	0.4
37	Specialized instruments	Manufacturing Instruments, Cultural and Office Machinery Specialized Instruments	38	0.4
38	Communication equipment	Manufacturing Communication Equipment, Computers and Other Electronic Equipment Communication Equipment Manufacturing	36	0.4
39	Travel services	Leasing and Commercial Services Business Services Travel Agencies	35	0.4
40	Other warehousing	Transport, Storage and Postal Services Warehousing Other Warehousing	33	0.3
Other categories (92 categories)			827	8.8

Notes: The table lists the 40 most frequent industry categories in the GGF investment sample (9,431 observations), which together account for 91.2% of total investments. Remaining low-frequency categories are grouped as “Other categories.” Industry labels follow the official Chinese National Economic Industry Classification (GB/T 4754).

Table A17: IO Spillover Robustness Across Top-Journal Style Specifications

Specification	ln(Hiring)			ln(Postings)		
	BL	FL	TE	BL	FL	TE
Baseline: firm FE + province×year FE	0.001 (0.048)	0.260*** (0.059)	−0.063 (0.043)	0.005 (0.049)	0.256*** (0.061)	−0.075 (0.040)
Weighted linkage + firm FE + city×year FE	−0.018 (0.062)	0.021 (0.041)	−0.005 (0.040)	−0.021 (0.062)	0.019 (0.040)	−0.018 (0.037)
Exclude own sector + firm FE + city×year FE	−0.048 (0.048)	0.088** (0.039)	−0.026 (0.045)	−0.049 (0.048)	0.086** (0.037)	−0.038 (0.041)
Exclude own sector (weighted) + firm FE + city×year FE	0.001 (0.053)	0.050** (0.021)	0.000 (0.043)	−0.003 (0.054)	0.047** (0.018)	−0.013 (0.040)

Notes: This table reports ATE coefficients from continuous IO-linkage spillover regressions on untreated firms in exposed cities. BL = backward linkage, FL = forward linkage, TE = same-city same-sector treated exposure control. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Spillover Effects with Staggered-Timing Robust Estimators (SA/CS)

	SA (IW Event Study)	CS (ATT _{<i>g,t</i>})
<i>Panel A: Quantity</i>		
ln(Hiring)	0.021 (0.032)	-0.104 (0.097)
ln(Posts)	0.023 (0.029)	-0.070 (0.091)
<i>Panel B: Price and Composition</i>		
ln(Max Wage)	0.051 (0.010)	0.044 (0.048)
High-skill share	0.019 (0.003)	-0.015 (0.024)
Reference period	$k = -1$	Pre-treatment average
Observations	141,135	sampled panel

Spillover windows aligned to the TWFE spillover tables by outcome.

CS uses `mod(firm_id,20)==0` sample for runtime and pair-balanced ATT aggregation.

Standard errors in parentheses.

Notes: Sun–Abraham and Callaway–Sant’Anna estimators are reported for the spillover design to address concerns about TWFE weighting under staggered adoption. Their near-zero estimates reinforce the paper’s revised interpretation that quantity spillovers are weak and not robust headline findings. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Tax-Sample Heterogeneity by Pre-Treatment Leverage

	(1) ln(Employment)	(2) ln(Wage p.c.)
GGF Investment	0.250*** (0.082)	0.222** (0.106)
× High pre-leverage	-0.004 (0.105)	-0.115 (0.142)
Observations	1,482,213	1,475,836
Firm FE	✓	✓
Province×Year FE	✓	✓

Notes: The table reports heterogeneity in tax-sample treatment effects by pre-treatment leverage groups. Fixed effects and clustering follow the baseline tax specification. Sample includes treated firms with non-missing pre-leverage. Firm and province×year fixed effects included. Standard errors clustered at the firm level in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes on city-level HHI analysis. We treat the HHI exercise as descriptive evidence on recruitment concentration within city-skill markets, rather than as a stand-alone welfare test of labor-market power.

Table A20: City-Level Labor-Market Concentration (HHI) After GGF Exposure

	(1) High-skill HHI	(2) Low-skill HHI
GGF city exposure (post)	0.015* (0.008)	0.007 (0.009)
City-industry-skill-years	19747	19747
Pretrend joint p-value	0.684	0.980
Post joint p-value	0.001	0.941
Min market posts	10	10
Min firms per market	4	4
Weighted	No	No
Event binning	Tail bins (<=-3, >=3)	Tail bins (<=-3, >=3)
Sample years	Keep all years	Keep all years
Segment pair	Tech vs Non-tech	Tech vs Non-tech
City sample	All cities	All cities

Standard errors in parentheses; SE clustered at city level.

FE: city-industry-skill FE + year-skill FE.

Strict constraints satisfied: 1; relaxed constraints satisfied: 1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: HHI is measured from vacancy shares within city-year-skill cells. Regressions include city and year fixed effects with city-clustered standard errors; the estimates are descriptive evidence on recruitment concentration, not a stand-alone welfare analysis of local labor-market power. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes on city-industry reallocation analysis. These appendix GE materials use a vacancy-based city×industry×year panel; the parallel tax city-industry aggregation is too sparse to serve as a stable benchmark.

Table A21: City-Industry Reallocation in Vacancy Demand

	Vacancy target	Vacancy other
Post-city exposure	2.659*** (0.312)	0.108 (0.078)
Observations	414	8,585
City×Industry FE	Yes	Yes
Province×Year FE	Yes	Yes

Notes: The city-industry panel is built from vacancy microdata. “Target” cells are industries that ever directly receive GGF investment in a city; “Other” cells are remaining mapped industries in the same exposed cities. Regressions include city-by-industry and province-by-year fixed effects with standard errors clustered at the city level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes on long-window tax event study. The long-window tax tables confirm the paper’s main

tax interpretation: employment shows pre-trend problems consistent with selection-on-dip and rebound, while wage per worker remains the cleaner tax outcome.

Table A22: Long-Window Tax Event Study in the Matched Clean-Map Sample

	ln(Employment)	ln(Wage p.c.)
t<=-5	-0.517*** (0.162)	-0.032 (0.075)
t=-4	-0.246 (0.175)	-0.081 (0.072)
t=-3	-0.233 (0.158)	-0.071 (0.071)
t=-2	-0.114 (0.135)	-0.124* (0.074)
t=0	0.253* (0.135)	0.135** (0.066)
t=1	0.435*** (0.155)	0.124 (0.076)
t=2	0.197 (0.172)	0.144* (0.085)
t=3	0.375** (0.173)	0.021 (0.095)
t=4	0.291 (0.222)	0.033 (0.107)
t>=5	0.069 (0.227)	-0.052 (0.110)

Notes: The matched clean-map tax sample extends the event-study window to five years before and after treatment. The long-window employment series shows a pronounced early dip before treatment, which is why matched-sample tax employment is described as supportive rather than fully causal. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Pre-Trend and Post-Treatment Joint Tests for the Matched Tax Event Study

Outcome	Pre-trend p -value	Post-treatment p -value	Observations
ln(Employment)	0.023	0.044	21,796
ln(Wage p.c.)	0.491	0.282	17,463

Notes: This table summarizes joint significance tests for the matched long-window tax event study. The employment pre-trend rejection is central to the paper's revised interpretation of tax employment as directional evidence, while wage per worker remains the cleaner tax outcome. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A24: Long-Window Tax Event Study in the Full Clean-Map Sample

	ln(Employment)	ln(Wage p.c.)
t≤-5	-0.695*** (0.168)	-0.114 (0.075)
t=-4	-0.351** (0.174)	-0.128* (0.074)
t=-3	-0.200 (0.173)	-0.122* (0.074)
t=-2	-0.075 (0.148)	-0.141* (0.077)
t=0	0.157 (0.140)	0.094 (0.073)
t=1	0.332** (0.156)	0.052 (0.078)
t=2	0.421*** (0.160)	0.131* (0.079)
t=3	0.370** (0.162)	0.052 (0.086)
t=4	0.296 (0.183)	0.205** (0.093)
t≥5	0.285 (0.191)	0.111 (0.091)

Notes: The full clean-map tax sample provides a broader but noisier long-window comparison set. As in the matched sample, the employment dynamics are compatible with selection-on-dip and rebound, so these estimates are retained for transparency rather than as primary causal evidence. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A25: Pre-Trend and Post-Treatment Joint Tests for the Full Clean-Map Tax Event Study

Outcome	Pre-trend p -value	Post-treatment p -value	Observations
ln(Employment)	0.000	0.260	5,283,987
ln(Wage p.c.)	0.328	0.212	4,006,480

Notes: Joint tests for the full clean-map tax sample confirm that employment fails the pre-trend requirement even more strongly than in the matched sample. These diagnostics justify the paper's decision to demote tax employment from validation to supportive corroboration. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes on fresh-graduate vacancy analysis. The fresh-graduate sample is included as external-platform corroboration of the direct posting effect, not as a separate headline validation design.

Table A26: Direct Effects in the Fresh-Graduate Vacancy Sample

Outcome	Coefficient	Std. Error
ln(Freshgrad Posts)	0.121***	(0.033)
ln(Freshgrad Max Wage)	0.001	(0.030)
ln(Freshgrad Hires)	0.016	(0.051)
Bachelor+ Share	0.007	(0.017)

Notes: This table uses an independent fresh-graduate vacancy dataset to check external-platform consistency. The positive posting response serves as corroborating evidence that the direct vacancy effect is not unique to the main recruitment platform. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A27: [Appendix] IO-Based Spillover ATE: Same vs. Unsame Industries

	Same Industry	Upstream	Downstream	Unrelated
ln(Hiring+1)	0.377** (0.181)	0.067*** (0.022)	0.222 (0.244)	0.222 (0.244)
ln(Posts+1)	0.392*** (0.040)	0.178* (0.088)	0.205 (0.236)	0.205 (0.236)
ln(Max Wage+1)	0.125*** (0.024)	-0.157 (0.362)	0.078 (0.063)	0.078 (0.063)
ln(Min Wage+1)	0.152*** (0.031)	-0.062 (0.419)	0.111* (0.062)	0.111* (0.062)
High-skill share	0.095*** (0.023)	-0.071 (0.090)	-0.028 (0.042)	-0.028 (0.042)
Tech-post share	0.042*** (0.008)	0.006 (0.009)	0.025 (0.015)	0.025 (0.015)
Observations (max)	2,538	2,538	2,538	2,538
Firm FE	.	.	.	.
Year FE	.	.	.	.
Cluster	City	City	City	City

Notes: The table reports spillover ATE regressions for never-treated firms with matched IO42 codes, by city-level IO relationship group. Estimates come from a pooled interaction specification with Unrelated as the baseline group; coefficients for Same/Upstream/Downstream are computed as the sum of the baseline treated_post coefficient and the corresponding interaction term (lincom), with standard errors from lincom. The primary specification absorbs firm and province-by-year fixed effects; when province-by-year mixed support is weak, it falls back to firm-by-year fixed effects. Standard errors are clustered at the city level. The analysis is restricted to firms with matched IO42 industry codes (N=183,129 firms, 10.4% of the full spillover sample).

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.